SHEAVES ON LOCALLY COMPACT SPACES

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Fix a coefficient ring k, which we assume to be commutative and Noetherian.

1. Locally compact spaces

Definition 1.1. Let X be a topological space. We say that X is *locally compact* if:

(1) X is Hausdorff;
(2) For every point $x \in X$ and every open neighbourhood U of x, there exists a smaller open neighbourhood V of x such that $\overline{V} \subset U$ and $\overline{V}$ is compact.

Euclidean space $\mathbb{R}^n$ is locally compact. More generally, a topological manifold is locally compact.

Proposition 1.2. *Suppose X is a locally compact space, and $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf. For every compact subset K of X, restriction induces a canonical isomorphism*

$$\varinjlim \Gamma(U, \mathcal{F}) \cong \Gamma(K, \mathcal{F})$$

where U ranges over the open supersets of K.

Proof. Let $i : K \to X$ denote the inclusion. Since the functors i^* and i_* form an adjoint pair, we have a unit natural transformation $\mathrm{id} \to i_* \circ i^*$, and hence a map $\mathcal{F} \to i_* i^* \mathcal{F}$. This map of sheaves gives rise to a k-linear map

$$\Gamma(U, \mathcal{F}) \longrightarrow \Gamma(K, i^*\mathcal{F}) =: \Gamma(K, \mathcal{F})$$

for every open superset U of K; moreover, if $U \supset U'$ are nested open sets containing K, then the triangle

$$\begin{tikzcd} \Gamma(U, \mathcal{F}) \arrow[rr] \arrow[dr] & & \Gamma(U', \mathcal{F}) \arrow[dl] \\ & \Gamma(K, \mathcal{F}) & \end{tikzcd}$$

commutes. So the universal property gives a map $\varinjlim \Gamma(U, \mathcal{F}) \to \Gamma(K, \mathcal{F})$. To see that this map is injective, note that if $s \in \Gamma(U, \mathcal{F})$ is a section that restricts to zero on K, then for all $x \in K$, $s_x = 0$ in the stalk $\mathcal{F}_x = (\mathcal{F}|_K)_x$, so that there is an open set $V_x \subset U$ such that $s|_{V_x} = 0$. Setting $V := \cup_{x \in X} V_x$, we have $s|_V = 0$, which proves the class of s is zero in $\varinjlim \Gamma(U, \mathcal{F})$.

We now prove surjectivity. Recall that $\mathcal{F}|_K = i^*\mathcal{F}$ is the sheaf associated to the presheaf on K:

$$V \mapsto \varinjlim_{U \supset V} \mathcal{F}(U);$$

here V is an open subset of K, and U ranges over the supersets of V which are open in X. So if $t \in \Gamma(K, \mathcal{F})$ is a section, then there exists an open cover $\{U_i\}_{i \in I}$ of K and sections $s_i \in \Gamma(U_i, \mathcal{F})$ such that $(s_i)|_{U_i \cap K} = t|_{U_i \cap K}$. Since each x is in some U_i, the fact that X is locally compact implies there exists a smaller neighbourhood V_i of

x such that $\overline{V_i} \subset U_i$. Then $\{V_i\}_{i\in I}$ is an open cover of K, which is compact, so we may assume that I is finite. For each $x \in X$, let $I(x) := \{i \in I \mid x \in \overline{V_i}\}$, and define

$$W := \{x \in \cup_{i\in I} V_i \mid (s_i)_x = (s_j)_x \text{ for all } i, j \in I(x)\}.$$

Note that $K \subset W$. We claim that W is an open set. Given $x \in W$, let

$$V_x := \bigcap_{i\in I,\, x\in V_i} V_i \cap \bigcap_{j\in I\setminus I(x)} \overline{V_j}^{\mathrm{c}}.$$

Then V_x is an open neighbourhood of x, and if $y \in V_x$ then the fact that $y \notin \overline{V_j}$ for all $j \notin I(x)$ implies $I(y) \subset I(x)$. Now if we fix $i, j \in I(x)$, then there is an open neighbourhood $W^{ij} \subset V_i \cap V_j$ of x such that $(s_i)|_{W^{ij}} = (s_j)|_{W^{ij}}$. The intersection

$$V_x \cap \bigcap_{i,j\in I(x)} W^{ij}$$

is an open neighbourhood of x, and it is contained in W. Hence, W is open in X. Now $\{V_i \cap W\}_{i\in I}$ is an open cover of W, and the sections $(s_i)|_{V_i\cap W}$ agree on the overlaps $V_i \cap V_j \cap W$ by the definition of W. Since $\mathcal{F}$ is a sheaf, these sections glue to a section $s \in \Gamma(W, \mathcal{F})$, which restricts to t on K. □

Fix locally compact spaces X, Y, and a continuous map $f : X \to Y$.

Definition 1.3. If $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf on X, then the *direct image with compact support* of $\mathcal{F}$ is the presheaf $f_!\mathcal{F}$ on Y defined by setting

$$\Gamma(U, f_!\mathcal{F}) := \{s \in \Gamma(f^{-1}(U), \mathcal{F}) \mid f : \mathrm{Supp}(s) \to U \text{ is proper}\},$$

for every open set $U \subset Y$.

Lemma 1.4. *The presheaf $f_!\mathcal{F}$ is a sheaf.*

Proof. Since $f_!\mathcal{F}$ is a subpresheaf of $f_*\mathcal{F}$, any set of compatible sections of $f_!\mathcal{F}$ can be glued uniquely to a section of $f_*\mathcal{F}$, and all we have to check is that this section belongs to $f_!\mathcal{F}$. So suppose we are given an open set $U \subset Y$, a section $s \in \Gamma(f^{-1}(U), \mathcal{F})$, and an open cover $\{U_i\}_{i\in I}$ of U. Write $S := \mathrm{Supp}(s)$ and $S_i := S \cap f^{-1}(U_i)$. What we have to show is that if $f|_{S_i} : S_i \to U_i$ is proper for each $i \in I$, then $f|_S : S \to U$ is proper.

Let $K \subset U$ be a compact set. Since Y is locally compact, and every point $x \in K$ is contained in some U_i, there is a compact set K_i' such that $K_i' \subset U_i$, and the interior of K_i' is an open neighbourhood of x. By compactness of K, there is a finite subset J of I such that $K \subset \cup_{j\in J} K_j'$. Then $K_j := K_j' \cap K$ is compact because it is a closed subset of a compact set (recall that compact subspaces of Hausdorff spaces are closed), so that we have $K = \cup_{j\in J} K_j$.

We must show that $f^{-1}(K) \cap S$ is compact. But using the fact that $K = \cup_{j\in J} K_j$, we can write it as a finite union of compact sets:

$$\begin{aligned} f^{-1}(K) \cap S &= \bigcup_{j\in J} f^{-1}(K_j) \cap S \\ &= \bigcup_{j\in J} f^{-1}(K_j) \cap S_j, \end{aligned}$$

where the compactness of $f^{-1}(K_j) \cap S_j$ follows from the fact that $K_j \subset U_j$ is compact and our assumption that $f|_{S_i} : S_i \to U_i$ is proper. □

In fact, we have a functor

$$f_! : \mathbf{Sh}(X, k) \longrightarrow \mathbf{Sh}(Y, k),$$

which is a subfunctor of f_*; this follows from the fact that if $\alpha : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves and s is a local section of $\mathcal{F}$, then $\mathrm{Supp}(\alpha(s)) \subset \mathrm{Supp}(s)$. Note that if f is proper, then $f_! = f_*$. Like f_*, the functor $f_!$ is left exact, so it has a right derived functor:

$$Rf_! : \mathrm{D}^+(X, k) \longrightarrow \mathrm{D}^+(Y, k).$$

The case where Y is a point deserves a special notation:

Definition 1.5. If $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf on X, then we set

$$\Gamma_{\mathrm{c}}(X, \mathcal{F}) := \{s \in \Gamma(U, \mathcal{F}) \mid \mathrm{Supp}(s) \text{ is compact}\}.$$

This defines a functor $\Gamma_{\mathrm{c}}(X, -) : \mathbf{Sh}(X, k) \to k{-}\mathbf{Mod}$. The higher derived functors of $\Gamma_{\mathrm{c}}(X, -)$ have their own special notation: we set

$$H^i_{\mathrm{c}}(X, -) := R^i\Gamma_{\mathrm{c}}(X, -),$$

and refer to these functors as *sheaf cohomology with compact support.*

We now study the special case where $j : U \hookrightarrow X$ is the inclusion of an open subset. In this situation, the functor $j_!$ is known as *extension by zero.* Unpacking the definition, we find that

$$\Gamma(V, j_!\mathcal{G}) = \{s \in \Gamma(U \cap V, \mathcal{G}) \mid \mathrm{Supp}(s) \text{ is closed in } V\};$$

here $\mathcal{G} \in \mathbf{Sh}(U, k)$ and $V \subset X$ is open. It follows that if $x \in X$, we have

$$(j_!\mathcal{G})_x = \begin{cases} \mathcal{G}_x & \text{if } x \in U \\ 0 & \text{if } x \notin U. \end{cases} \tag{1.1}$$

This shows that $j_!$ is exact, and that the image of $j_!$ consists of sheaves on X supported in U. In fact, the functors $j_!$ and j^* define an equivalence of categories

$$\mathbf{Sh}(U, k) \underset{j^*}{\overset{j_!}{\rightleftarrows}} \{\mathcal{F} \in \mathbf{Sh}(X, k) \mid \mathrm{Supp}(\mathcal{F}) \subset U\},$$

where the category on the right is by definition a full subcategory of $\mathbf{Sh}(X, k)$. To see this, note that if $\mathcal{G} \in \mathbf{Sh}(U, k)$, then we have an equality $j^*j_!\mathcal{G} = \mathcal{G}$. On the other hand, if $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf such that $\mathrm{Supp}(\mathcal{F}) \subset U$, then one easily sees by checking the stalks that the natural map $\mathcal{F} \to j_!j^*\mathcal{F}$ is an isomorphism.

Proposition 1.6. *The functor $j_!$ is left adjoint to the restriction j^*.*

Proof. We first define a natural transformation $\varepsilon : j_!j^* \to \mathrm{id}$, which will be the counit of the adjunction. Given a sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$ and an open subset $V \subset X$, let $s \in \Gamma(V, j_!j^*\mathcal{F})$. Then s is a section of $\mathcal{F}$ over $U \cap V$ whose support $\mathrm{Supp}(s) \subset U \cap V$ is closed in V. Then V has a covering by two open sets:

$$V = (U \cap V) \cup (V \setminus \mathrm{Supp}(s)),$$

and since s and the zero section $0 \in \Gamma(V \setminus \mathrm{Supp}(s), \mathcal{F})$ agree on the overlap, they glue to a section $\varepsilon_{\mathcal{F}}(s) \in \Gamma(V, \mathcal{F})$. This defines $\varepsilon_{\mathcal{F}}$, and hence ε.

To show ε is the counit of an adjunction between $j_!$ and j^*, we have to verify that it satisfies the following universal property: for every map $\alpha : j_!\mathcal{G} \to \mathcal{F}$, where $\mathcal{G} \in \mathbf{Sh}(U, k)$, there exists a unique map $\beta : \mathcal{G} \to j^*\mathcal{F}$ making the diagram

$$\begin{tikzcd} j_!j^*\mathcal{F} \arrow[r, "\varepsilon_{\mathcal{F}}"] & \mathcal{F} \\ & j_!\mathcal{G} \arrow[ul, "j_!\beta"] \arrow[u, "\alpha"'] \end{tikzcd}$$

commute. Now if $W \subset U$ is open then $\mathcal{G}(W) = (j_!\mathcal{G})(W)$ and $(j_!j^*\mathcal{F})(W) = \mathcal{F}(W)$, so we may define $\beta(W) := \alpha(W)$. The commutativity of the diagram, and the uniqueness of β, can easily be checked on the stalks. $\square$

2. Soft sheaves

In studying the derived functors $Rf_!$ and $R\Gamma_c$, the following class of sheaves will turn out to be very useful:

Definition 2.1. We say that a sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$ is *soft* if for any compact subset $K \subset X$, the restriction map $\Gamma(X, \mathcal{F}) \to \Gamma(K, \mathcal{F})$ is surjective.

It follows from Proposition 1.2 that flasque sheaves are soft; in particular, injective sheaves are soft. We will see more examples of soft sheaves in Section 4.

Proposition 2.2. *Suppose W is a locally closed subspace of X. If $\mathcal{F} \in \mathbf{Sh}(X, k)$ is soft, then $\mathcal{F}|_W \in \mathbf{Sh}(W, k)$ is soft.*

Proof. This is immediate from the definition. $\square$

Recall that a sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$ is *$f_!$-acyclic* if $R^i f_!\mathcal{F} = 0$ for all $i > 0$. Our main goal in this section is to prove that soft sheaves are $f_!$-acyclic.

Lemma 2.3. *Let $W_2 \subset W_1 \subset X$ be nested subspaces of X, and let $i_1 : W_1 \to X$ and $i_2 : W_2 \to X$ denote the inclusion maps. For any sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$, we have a natural morphism $(i_1)_*(i_1)^*\mathcal{F} \to (i_2)_*(i_2)^*\mathcal{F}$.*

Proof. Let $j : W_2 \to W_1$ denote the inclusion, so that $i_2 = i_1 \circ j$. The component of the unit natural transformation $\mathrm{id} \to j_*j^*$ at the sheaf $(i_1)^*\mathcal{F}$ is a morphism of sheaves on W_1:

$$(i_1)^*\mathcal{F} \longrightarrow j_*j^*\big((i_1)^*\mathcal{F}\big) = j_* \circ (i_2)^*\mathcal{F}.$$

Applying the functor $(i_1)_*$ then gives the desired morphism of sheaves on X. $\square$

We are used to gluing together sections of sheaves over open subsets. The next lemma says that we can do the same for closed subsets.

Lemma 2.4. *Suppose A and B are closed subsets of X. If $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf and $a \in \Gamma(A, \mathcal{F})$, $b \in \Gamma(B, \mathcal{F})$ are sections such that $a|_{A\cap B} = b|_{A\cap B}$, then there exists a unique section $s \in \Gamma(A \cup B, \mathcal{F})$ such that $s|_A = a$ and $s|_B = b$.*

Proof. Let $h : A \cup B \to X$, $i : A \to X$, $j : B \to X$, $k : A \cap B \to X$ denote the inclusions. Consider the following sequence of sheaves on X:

$$0 \longrightarrow h_*h^*\mathcal{F} \longrightarrow i_*i^*\mathcal{F} \oplus j_*j^*\mathcal{F} \longrightarrow k_*k^*\mathcal{F} \longrightarrow 0, \tag{2.1}$$

where the morphisms are sums and differences of the maps from Lemma 2.3. By checking the stalks, we see that this sequence is exact (this is where we use the fact that A and B are closed). Taking global sections, we get an exact sequence of k-modules

$$0 \longrightarrow \Gamma(A \cup B, \mathcal{F}) \longrightarrow \Gamma(A, \mathcal{F}) \oplus \Gamma(B, \mathcal{F}) \longrightarrow \Gamma(A \cap B, \mathcal{F}).$$

Since (a, b) is in the kernel of the second map, exactness implies the existence of a unique extension s over $A \cup B$. $\square$

Proposition 2.5. *Let $\mathcal{F} \in \mathbf{Sh}(X, k)$ a soft sheaf. Then $\mathcal{F}$ is soft if and only if for each closed subset $Z \subset X$, the restriction map $\Gamma_{\mathrm{c}}(X, \mathcal{F}) \to \Gamma_{\mathrm{c}}(Z, \mathcal{F})$ is surjective.*

Proof. One direction is clear, so assume $\mathcal{F}$ is soft. Let $s \in \Gamma_{\mathrm{c}}(Z, \mathcal{F})$. Choose a compact set $L \subset X$ whose interior L° contains the support of s (this is possible since $\mathrm{Supp}(s)$ is compact). By Lemma 2.4, the sections $s|_L \in \Gamma(L \cap Z, \mathcal{F})$ and $0 \in \Gamma(\partial L, \mathcal{F})$ glue to a section over $(L \cap Z) \cup \partial L$, which we will continue to denote by s. Now because $\mathcal{F}|_L$ is soft and $(L \cap Z) \cup \partial L$ is compact, there exists a section $t \in \Gamma(L, \mathcal{F})$ that restricts to s on $(L \cap Z) \cup \partial L$. Finally, because the section t is zero on the boundary ∂L, we can extend it by zero to a global section of $\mathcal{F}$ over X (use the open cover L° and $X \setminus \mathrm{Supp}(t)$). This section restricts on Z to our original section s. $\square$

Lemma 2.6. *For each point $y \in Y$, denote the fibre $f^{-1}(y)$ by X_y. Then for every sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$, we have a canonical isomorphism*

$$(f_! \mathcal{F})_y \cong \Gamma_{\mathrm{c}}(X_y, \mathcal{F}|_{X_y}),$$

which is natural with respect to $\mathcal{F}$.

Proof. We begin by defining a map $(f_! \mathcal{F})_y \to \Gamma_{\mathrm{c}}(X_y, \mathcal{F}|_{X_y})$. Since

$$(f_! \mathcal{F})_y = \varinjlim_{V \ni y} \Gamma(V, f_! \mathcal{F}),$$

this is equivalent to defining a system of maps $\Gamma(V, f_! \mathcal{F}) \to \Gamma_{\mathrm{c}}(X_y, \mathcal{F}|_{X_y})$, where V varies over all the open neighbourhoods of y. Now given a section of $f_! \mathcal{F}$ over some such V, then the restriction of the corresponding section $s \in \Gamma(f^{-1}(V), \mathcal{F})$ to X_y has compact support; to see this, note that the outer square of the diagram

$$\begin{array}{ccc} \mathrm{Supp}(s|_{X_y}) & \longrightarrow & \mathrm{Supp}(s) \\ \downarrow & & \downarrow \\ X_y & \longrightarrow & f^{-1}(V) \\ \downarrow & & \downarrow \\ \{y\} & \longrightarrow & V \end{array}$$

is Cartesian, and use the fact that properness is preserved under base change. Hence, the restriction maps $\Gamma(V, f_! \mathcal{F}) \to \Gamma(X_y, \mathcal{F}|_{X_y})$ factor through $\Gamma_{\mathrm{c}}(X_y, \mathcal{F}|_{X_y})$, and they define the required map.

To show this map is injective, let V be an open neighbourhood of y, and consider a section $s \in \Gamma(V, f_! \mathcal{F})$ whose restriction to the fibre X_y is zero. This means that $\mathrm{Supp}(s) \cap X_y = \varnothing$, or equivalently, that $y \notin f(\mathrm{Supp}(s))$. Now since $f|_{\mathrm{Supp}(s)}$ is a proper map, it is closed, so $f(\mathrm{Supp}(s))$ is closed in V. If we let W denote the open complement $V \setminus f(\mathrm{Supp}(s))$, then W is an open neighbourhood of y such that $s|_W = 0$, which shows $s_y = 0$ in the stalk $(f_! \mathcal{F})_y$.

We now prove surjectivity. In the case where $\mathcal{F}$ is soft, this follows from the commutative diagram

$$\begin{tikzcd} \Gamma_c(X,\mathcal{F}) \arrow[r, two heads] \arrow[d] & \Gamma_c(X_y,\mathcal{F}) \\ \Gamma_c(f^{-1}(V),\mathcal{F}) \arrow[r, hook] & \Gamma(V, f_!\mathcal{F}), \arrow[u] \end{tikzcd}$$

where the top horizontal arrow is surjective by Proposition 2.5. In the general case, choose injective sheaves $\mathcal{I}$ and $\mathcal{J}$ so that we have an exact sequence $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{J}$. Then we have the following the commutative diagram with exact rows:

$$\begin{tikzcd} 0 \arrow[r] & (f_!\mathcal{F})_y \arrow[r] \arrow[d] & (f_!\mathcal{I})_y \arrow[r] \arrow[d] & (f_!\mathcal{J})_y \arrow[d] \\ 0 \arrow[r] & \Gamma(X_y, \mathcal{F}|_{X_y}) \arrow[r] & \Gamma(X_y, \mathcal{I}|_{X_y}) \arrow[r] & \Gamma(X_y, \mathcal{J}|_{X_y}). \end{tikzcd}$$

Since $\mathcal{I}$ and $\mathcal{J}$ are soft, the middle and right vertical maps are isomorphisms by what we have already shown, so the left map must be as well. $\square$

Lemma 2.7. *If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is a short exact sequence in $\mathbf{Sh}(X,k)$ such that $\mathcal{F}'$ is soft, then the sequence $0 \to f_!\mathcal{F}' \to f_!\mathcal{F} \to f_!\mathcal{F}'' \to 0$ is exact in $\mathbf{Sh}(Y,k)$.*

Proof. Let $y \in Y$. By Lemma 2.6, we have a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & (f_!\mathcal{F}')_y \arrow[r] \arrow[d, "\sim"] & (f_!\mathcal{F})_y \arrow[r] \arrow[d, "\sim"] & (f_!\mathcal{F}'')_y \arrow[r] \arrow[d, "\sim"] & 0 \\ 0 \arrow[r] & \Gamma_c(X_y, \mathcal{F}'|_{X_y}) \arrow[r] & \Gamma_c(X_y, \mathcal{F}|_{X_y}) \arrow[r] & \Gamma_c(X_y, \mathcal{F}''|_{X_y}) \arrow[r] & 0. \end{tikzcd}$$

Since $\mathcal{F}'|_{X_y}$ is soft, it is enough to prove that the sequence

$$0 \longrightarrow \Gamma_c(X,\mathcal{F}') \longrightarrow \Gamma_c(X,\mathcal{F}) \longrightarrow \Gamma_c(X,\mathcal{F}'') \longrightarrow 0$$

is exact.

We first prove this assuming X is compact. Let $t \in \Gamma_c(X,\mathcal{F}'') = \Gamma(X,\mathcal{F}'')$. Since the map $\mathcal{F} \to \mathcal{F}''$ is surjective, there exists a finite compact cover $K_1, \dots, K_n$ of X and sections $s_i \in \Gamma(K_i, \mathcal{F})$ that lift $t|_{K_i}$, such that the interiors of the K_i cover X. To patch the s_i together into a global section which lifts t, we use induction on n, and it is enough to prove the case $n = 2$. Consider the section

$$e' := s_1|_{K_1 \cap K_2} - s_2|_{K_1 \cap K_2} \in \Gamma(K_1 \cap K_2, \mathcal{F}).$$

Since e' maps to zero in $\Gamma(K_1 \cap K_2, \mathcal{F}'')$, it belongs to $\Gamma(K_1 \cap K_2, \mathcal{F}')$. By softness of $\mathcal{F}'$, there exists a section $e \in \Gamma(X,\mathcal{F}')$ such that $e|_{K_1 \cap K_2} = e'$. The sections s_1 and $s_2 + e|_{K_2}$ agree on $K_1 \cap K_2$, so Lemma 2.4 implies that they glue to a global section s of $\mathcal{F}$. This section s maps to $t|_{K_i}$ when restricted to K_i, so by the uniqueness part of Lemma 2.4, it maps to t.

We now deal with the noncompact case. Let $t \in \Gamma_c(X,\mathcal{F}'')$. Choosing a compact subset $K \subset X$ which contains $\mathrm{Supp}(t)$ in its interior, the restriction $t|_K$ lifts to a section $s' \in \Gamma(K,\mathcal{F})$ by the previous paragraph. Now if the restriction of s to the boundary ∂K were zero, then we could extend it by zero to a compactly supported global section of $\mathcal{F}$. In the rest of the proof, we will see that we can reduce to this case. Note that since $\mathrm{Supp}(t) \cap \partial K = \varnothing$, $s|_{\partial K}$ maps to zero in $\Gamma(\partial K, \mathcal{F}'')$, so $s|_{\partial K}$

actually belongs to $\Gamma(\partial K, \mathcal{F}')$. Because $\mathcal{F}'$ is soft, s extends to a section $e \in \Gamma(K, \mathcal{F}')$. Then $s' := s - e$ maps to $t|_K$ and restricts to zero on ∂K, so it can be extended by zero to a global section of $\mathcal{F}$. □

Lemma 2.8. *If* $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ *be a short exact sequence in* $\mathbf{Sh}(X, k)$*. If* $\mathcal{F}$ *and* $\mathcal{F}'$ *are soft, then* $\mathcal{F}''$ *is soft too.*

Proof. Let $K \subset X$ be a compact subset, and consider the commutative square

$$\begin{array}{ccc} \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, \mathcal{F}'') \\ \downarrow & & \downarrow \\ \Gamma(K, \mathcal{F}) & \longrightarrow & \Gamma(K, \mathcal{F}''). \end{array}$$

Since $\mathcal{F}$ is soft, the left vertical map is surjective. By Lemma 2.7, the horizontal maps are surjective. Hence, $\Gamma(X, \mathcal{F}'') \to \Gamma(K, \mathcal{F}'')$ is surjective too, so $\mathcal{F}''$ is soft. □

Theorem 2.9. *Soft sheaves are acyclic for the functor* $f_!$*.*

Proof. Let $\mathcal{F} \in \mathbf{Sh}(X, k)$ be a soft sheaf. We will prove that $R^i f_!(\mathcal{F}) = 0$ for $i > 0$ using induction on i. Choose an embedding $\mathcal{F} \hookrightarrow \mathcal{I}$ where $\mathcal{I} \in \mathbf{Sh}(X, k)$ is injective, and consider the long exact sequence of the functor $f_!$ associated to the short exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0,$$

where $\mathcal{G}$ is the cokernel of the embedding. In particular, since $R^1 f_!(\mathcal{I}) = 0$ by the injectivity of $\mathcal{I}$, the sequence

$$f_!(\mathcal{I}) \longrightarrow f_!(\mathcal{G}) \longrightarrow R^1 f_!(\mathcal{F}) \longrightarrow 0$$

is exact; in other words, $R^1 f_!(\mathcal{F})$ is the cokernel of the map $f_!(\mathcal{I}) \to f_!(\mathcal{G})$. But $\mathcal{F}$ is soft, so Lemma 2.7 implies this map is an epimorphism, and hence $R^1 f_!(\mathcal{F}) = 0$. Now assume the result holds for $i > 0$. We know that $R^i f_!(\mathcal{I}) = R^{i+1} f_!(\mathcal{I}) = 0$, so we have an isomorphism $R^{i+1} f_!(\mathcal{F}) \cong R^i f_!(\mathcal{G})$. But the induction hypothesis applies to $\mathcal{G}$ since it is soft by Lemma 2.8, so $R^i f_!(\mathcal{G}) = 0$, and hence $R^{i+1} f_!(\mathcal{F}) = 0$. □

3. Compactly supported cohomology

We wish to show that compactly supported cohomology commutes with direct limits of sheaves. To spell this out, let $\{\mathcal{F}_\alpha\}_{\alpha \in A}$ be a direct system in $\mathbf{Sh}(X, k)$ over a directed set $(A, \leq)$, with transition maps $t_{\alpha,\beta} : \mathcal{F}_\alpha \to \mathcal{F}_\beta$ for $\alpha \leq \beta$. Note that if we apply the functor $H^i_c(X, -)$, we get a direct system of k-modules $\{H^i_c(X, \mathcal{F}_\alpha)\}_{\alpha \in A}$ over $(A, \leq)$. What we want to show is that the induced map

$$\varinjlim H^i_c(X, \mathcal{F}_\alpha) \longrightarrow H^i_c(X, \varinjlim \mathcal{F}_\alpha). \tag{3.1}$$

is an isomorphism.

Lemma 3.1. *The map* (3.1) *is an isomorphism when* $i = 0$*.*

Proof. For each point $x \in X$, the set of stalks $\{(\mathcal{F}_\alpha)_x\}_{\alpha \in A}$ is a direct system of k-modules over $(A, \leq)$. Observe that $\varinjlim (\mathcal{F}_\alpha)_x = (\varinjlim \mathcal{F}_\alpha)_x$, since taking the stalk at x can be interpreted as applying the functor j^*, where $j : \{x\} \hookrightarrow X$ denotes the inclusion, and as a left adjoint to j_*, j^* preserves all colimits. (In fact, colimits commute with pullbacks in general, for the same reason.) So for each $\alpha_0 \in A$, we have a commutative diagram

$$\begin{tikzcd} \Gamma_c(X, \mathcal{F}_{\alpha_0}) \arrow[r] \arrow[d] & \Gamma_c(X, \varinjlim \mathcal{F}_\alpha) \arrow[d] \\ (\mathcal{F}_{\alpha_0})_x \arrow[r] \arrow[dr] & (\varinjlim \mathcal{F}_\alpha)_x \arrow[d, "\sim"] \\ & \varinjlim (\mathcal{F}_\alpha)_x \end{tikzcd}$$

We first show that (3.1) is injective. Suppose $s \in \Gamma_c(X, \mathcal{F}_{\alpha_0})$ maps to 0 in $\Gamma_c(X, \varinjlim \mathcal{F}_\alpha)$. We need to show that there exists a $\beta \in A$ such that $t_{\alpha_0,\beta}(s) = 0$. Since the germ s_x maps to 0 in $\varinjlim(\mathcal{F}_\alpha)_x = (\varinjlim \mathcal{F}_\alpha)_x$, there exists a $\beta_x \in A$ and an open neighbourhood V_x of x such that $t_{\alpha_0,\beta_x}(s)|_{V_x} = 0$. But $\mathrm{Supp}(s)$ is compact, so there exist finite many points $x_1, \dots, x_n \in X$ such that $\mathrm{Supp}(s) \subset V_1 \cup \dots \cup V_n$. If we then choose a $\beta \in A$ larger than all of $\beta_1, \dots, \beta_n$, we have $t_{\alpha_0,\beta}(s) = 0$, as required.

Next, we prove that (3.1) is surjective when X is compact. Let $t \in \Gamma(X, \varinjlim \mathcal{F}_\alpha)$. By compactness of X, there exists a finite open cover of X:

$$X = V_1 \cup \dots \cup V_n,$$

and for all $1 \leq i \leq n$, an element $\beta_i \in A$ and a section $s_i \in \Gamma(V_i, \mathcal{F}_{\beta_i})$ such that the image of s_i in $\Gamma(V_i, \varinjlim \mathcal{F}_\alpha)$ equals the restriction of t to V_i. Here we have used the fact that $(\varinjlim \mathcal{F}_\alpha)|_{V_i} = \varinjlim((\mathcal{F}_\alpha)|_{V_i})$, which is true because direct limits commute with restrictions. Now by choosing a $\beta \in A$ larger than all the β_i, and then replacing s_i with $t_{\beta_i,\beta}(s)$, we can compare s_i and s_j on the intersection $V_i \cap V_j$. Since both sections must equal $t|_{V_i \cap V_j}$ in the limit, there exists a $\beta_{i,j} \in A$ such that $t_{\beta,\beta_{i,j}}(s_i|_{V_i \cap V_j}) = t_{\beta,\beta_{i,j}}(s_j|_{V_i \cap V_j})$. Choose $\gamma \in A$ such that $\gamma \geq \beta_{i,j}$ for all i, j, and then replace s_i with $t_{\beta,\gamma}(s_i)$. Since $\mathcal{F}_\gamma$ is a sheaf, the sections s_i glue to a global section $s \in \Gamma(X, \mathcal{F}_\gamma)$. By construction, the image of s in $\Gamma(X, \varinjlim \mathcal{F}_\alpha)$ is equal to t.

Finally, we prove (3.1) is surjective in general. Let $t \in \Gamma_c(X, \varinjlim \mathcal{F}_\alpha)$, and choose a compact subset $K \subset X$ that contains $\mathrm{Supp}(t)$ in its interior. Then $t|_K \in \Gamma(X, \varinjlim \mathcal{F}_\alpha)$, so by what we just proved, there exists a $\gamma \in A$ and a section $s \in \Gamma(K, \mathcal{F}_\gamma)$ that maps to $t|_K$ in the limit. Since t is zero on the boundary ∂K, we can assume the same is true for s: if necessary, choose $\gamma' \geq \gamma$ such that $t_{\gamma,\gamma'}(s)|_{\partial K} = 0$, and then replace s with $t_{\gamma,\gamma'}(s)$. Because $s|_{\partial K} = 0$, it can be extended by zero to a global section of $\mathcal{F}_{\gamma'}$, which we continue to label s. This s maps to t in the limit, as required. □

Lemma 3.2. *The direct limit of soft sheaves is soft.*

Proof. Let $K \subset X$ be a compact subset. By Lemma 3.1, we have a commutative diagram

$$\begin{tikzcd} \varinjlim \Gamma_c(X, \mathcal{F}_\alpha) \arrow[r, "\sim"] \arrow[d] & \Gamma_c(X, \varinjlim \mathcal{F}_\alpha) \arrow[d] \\ \varinjlim \Gamma_c(K, \mathcal{F}_\alpha) \arrow[r, "\sim"] & \Gamma_c(K, \varinjlim \mathcal{F}_\alpha). \end{tikzcd}$$

Since $\mathcal{F}_\alpha$ is soft for each $\alpha \in A$, the left vertical map is surjective, so the right vertical map is surjective as well. This shows $\varinjlim \mathcal{F}_\alpha$ is soft, as required. □

Theorem 3.3. *The compactly supported cohomology functors $H^i_c(X,-)$ commute with direct limits of sheaves, i.e.* (3.1) *is an isomorphism for all $i \geq 0$.*

Proof. Let $G^\cdot_\alpha$ be the Godement resolution of $\mathcal{F}_\alpha$; for the details of the construction, see the proof of Proposition 6.4. Since $G^\cdot_\alpha$ is a flasque (hence soft) resolution of $\mathcal{F}_\alpha$, we have

$$H^i_c(X,\mathcal{F}_\alpha) \cong H^i(\Gamma_c(X,G^\cdot_\alpha)).$$

Now the Godement resolution is functorial in the sheaf it resolves, so $\{G^\cdot_\alpha\}_{\alpha\in A}$ is a direct system. By Lemma 3.2, $\varinjlim G^\cdot_\alpha$ is a soft sheaf, and it resolves $\varinjlim \mathcal{F}_\alpha$, since $\varinjlim$ preserves exactness. So we have

$$H^i_c(X,\varinjlim \mathcal{F}_\alpha) \cong H^i(\Gamma_c(X,\varinjlim G^\cdot_\alpha)).$$

Finally, it follows from Lemma 3.1 that

$$\varinjlim H^i(\Gamma_c(X,G^\cdot_\alpha)) \cong H^i(\varinjlim \Gamma_c(X,G^\cdot_\alpha)) \cong H^i(\Gamma_c(X,\varinjlim G^\cdot_\alpha)).$$

□

As an example of a directed set, consider a collection $\mathcal{C}$ of closed subsets of X, partially ordered so that $A \leq B \iff B \subset A$ for all $A, B \in \mathcal{C}$. In this instance, 'directed' means that for every two elements $A, B \in \mathcal{C}$ there exists a $C \in \mathcal{C}$ such that $C \subset A \cap B$. For any sheaf $\mathcal{F} \in \mathbf{Sh}(X,k)$, the sheaves $\{(i_Z)_*(i_Z)^*\mathcal{F}\}_{Z\in\mathcal{C}}$ form a direct system over $\mathcal{C}$, with the transition maps given by Lemma 2.3.

Corollary 3.4. *Suppose $\mathcal{C}$ is a directed set of closed subsets of X, and $\mathcal{F}$ is a sheaf. Then with $W := \cap_{Z\in\mathcal{C}} Z$, there is an isomorphism*

$$\varinjlim_{Z\in\mathcal{C}} H^i_c(Z,\mathcal{F}) \xrightarrow{\sim} H^i_c(W,\mathcal{F})$$

induced by the restriction maps.

Proof. Observe that $\varinjlim_{Z\in\mathcal{C}} (i_Z)_*(i_Z)^*\mathcal{F} = (i_W)_*(i_W)^*\mathcal{F}$, and apply Theorem 3.3. □

Next, we want to prove a base change theorem for $f_!$. We begin with a special case of the main result:

Proposition 3.5. *For every sheaf $\mathcal{F} \in \mathbf{Sh}(X,k)$ and every integer $i \geq 0$, we have a canonical isomorphism*

$$(R^i f_! \mathcal{F})_y \cong H^i_c(X_y, \mathcal{F}|_{X_y}),$$

which is natural with respect to $\mathcal{F}$.

Proof. The case $i = 0$ is Lemma 2.6, so let $i > 0$. Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\cdot$. We then have

$$(R^i f_! \mathcal{F})_y = (\mathcal{H}^i(f_!\mathcal{I}^\cdot))_y \cong H^i((f_!\mathcal{I}^\cdot)_y),$$

since passing to the stalks commutes with taking cohomology. By Lemma 2.6, this is the ith cohomology of the complex $\Gamma_c(X_y, \mathcal{I}^\cdot|_{X_y})$. But $\mathcal{I}^\cdot|_{X_y}$ is a soft resolution of $\mathcal{F}|_{X_y}$, since the restriction of a soft sheaf to a closed subspace is soft. By Theorem 2.9, we have

$$H^i(\Gamma_c(X_y, \mathcal{I}^\cdot|_{X_y})) \cong H^i_c(X_y, \mathcal{F}|_{X_y}),$$

and this completes the proof. □

Theorem 3.6. *Suppose we have a Cartesian diagram of locally compact spaces:*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y. \end{tikzcd}$$

Then we have a natural isomorphism of functors $\mathrm{D}^+(X,k) \to \mathrm{D}^+(Y',k)$*:*

$$g^* \circ Rf_! \cong Rf'_! \circ (g')^*.$$

In particular, for each $i \geq 0$*, we have* $g^* \circ R^i f_! \cong R^i f'_! \circ (g')^*$*.*

Proof. As a first step, we will prove that there is a natural isomorphism α

$$g^* \circ f_! \xrightarrow{\sim} f'_! \circ (g')^*.$$

To define α, we will describe a natural transformation β

$$f_! \longrightarrow g_* \circ f'_! \circ (g')^*$$

and then invoke the adjointness of g^* and g_*. Let $\mathcal{F} \in \mathbf{Sh}(X,k)$ be a sheaf, U an open subset of Y, and suppose $s \in \Gamma(U, f_!\mathcal{F})$. By definition, s is a section of $\mathcal{F}$ over $f^{-1}(U)$ such that $f|_{\mathrm{Supp}(s)}$ is a proper map. Consider the pullback $s' := (g')^*(s)$, which is a section of $(g')^*\mathcal{F}$ over $(g')^{-1}(f^{-1}(U)) = (f')^{-1}(g^{-1}(U))$. Now properness is preserved by base change, so the fact that the diagram

$$\begin{tikzcd} \mathrm{Supp}(s') \arrow[r] \arrow[d, "f'"] & \mathrm{Supp}(s) \arrow[d, "f"] \\ g^{-1}(U) \arrow[r] & U \end{tikzcd}$$

is Cartesian implies $f'|_{\mathrm{Supp}(s')}$ is proper. Hence, s' is a section of $f'_!(g')^*\mathcal{F}$ over $g^{-1}(U)$; that is, $s' \in \Gamma(U, g_* f'_!(g')^*\mathcal{F})$. The assignment $s \mapsto s'$ defines a morphism of sheaves $f_!\mathcal{F} \to g_* f'_!(g')^*\mathcal{F}$, which is clearly natural in $\mathcal{F}$. This defines the desired natural transformation β. Now letting $\varepsilon : g^* \circ g_* \to \mathrm{id}$ denote the counit of adjunction, we define α as the composition

$$g^* \circ f_! \xrightarrow{\mathrm{id}_{g^*} \circ \beta} g^* \circ g_* \circ f'_! \circ (g')^* \xrightarrow{\varepsilon \circ \mathrm{id}_{f'_! \circ (g')^*}} f'_! \circ (g')^*.$$

It remains to check that α is an isomorphism of functors. Let $y' \in Y'$ be a point. We claim that

$$(\alpha_{\mathcal{F}})_{y'} : (g^* f_! \mathcal{F})_{y'} \longrightarrow (f'_!(g')^*\mathcal{F})_{y'}$$

is an isomorphism. Note that with $y := g(y')$, we have

$$(g^* f_! \mathcal{F})_{y'} \cong (f_!\mathcal{F})_y \cong \Gamma_{\mathrm{c}}(X_y, \mathcal{F}|_{X_y}) \tag{3.2}$$

by Lemma 2.6. On the other hand, the same lemma implies

$$(f'_!(g')^*\mathcal{F})_{y'} \cong \Gamma_{\mathrm{c}}\big(X'_{y'}, ((g')^*\mathcal{F})|_{X'_{y'}}\big). \tag{3.3}$$

Now it follows from the explicit construction of $X' = X \times_Y Y'$ as a locally compact space that g' restricts to a homeomorphism between the fibres $X'_{y'}$ and X_y, and we have

$$((g')^*\mathcal{F})|_{X'_{y'}} = (g'|_{X'_{y'}})^*(\mathcal{F}|_{X_y}).$$

We claim that the diagram

$$\begin{tikzcd} (g^* f_! \mathcal{F})_{y'} \arrow[r, "(3.2)"] \arrow[d, "(\alpha_{\mathcal{F}})_{y'}"] & \Gamma_c(X_y, \mathcal{F}|_{X_y}) \arrow[d, "\sim"] \\ (f'_! (g')^* \mathcal{F})_{y'} \arrow[r, "(3.3)"] & \Gamma_c\big(X'_{y'}, (g'|_{X'_{y'}})^* (\mathcal{F}|_{X_y})\big) \end{tikzcd}$$

is commutative. If we identify $(g^* f_! \mathcal{F})_{y'}$ with $(f_! \mathcal{F})_y$, then $(\alpha_{\mathcal{F}})_{y'}$ is given by

$$(f_! \mathcal{F})_y \xrightarrow{\beta_y} (g_* f'_! (g')^* \mathcal{F})_y \longrightarrow (f'_! (g')^* \mathcal{F})_{y'},$$

where the second map is the natural map on stalks induced by ε. But β was essentially given by the pullback $(g')^*$, so from here commutativity is easy to check. This completes the proof that α is a natural isomorphism.

On the level of derived functors, we have an induced natural isomorphism

$$R(g^* \circ f_!) \cong R(f'_! \circ (g')^*).$$

Since g^* is an exact functor, the theorem on compositions of total derived functors implies

$$R(g^* \circ f_!) \cong R(g^*) \circ Rf_! \cong g^* \circ Rf_!.$$

We now wish to apply the same theorem to $f'_! \circ (g')^*$ to conclude that

$$R(f'_! \circ (g')^*) \cong Rf'_! \circ (g')^*,$$

but to do so, we have to explain why the image of an injective sheaf $\mathcal{I} \in \mathbf{Sh}(X, k)$ under $(g')^*$ is $f'_!$-acyclic. Let $\mathcal{G} := (g')^* \mathcal{I}$. We claim that for each point $y' \in Y'$, the restriction of $\mathcal{G}$ to the fibre $X'_{y'} = (f')^{-1}(y')$ is soft. But if $y = g(y')$, then as above we have a homeomorphism $X'_{y'} \cong X_y$, and $\mathcal{G}|_{X'_{y'}}$ corresponds to $\mathcal{I}|_{X_y}$ under this identification. Therefore, $\mathcal{G}|_{X'_{y'}}$ is soft, and it follows from Proposition 3.5 that

$$(R^i f'_! \mathcal{G})_{y'} \cong H^i_c(X'_{y'}, \mathcal{G}|_{X'_{y'}}) = 0$$

for all $i > 0$, since soft sheaves are Γ_c-acyclic by Theorem 2.9. Therefore, $R^i f'_! \mathcal{G}$ must be the zero sheaf; that is, $\mathcal{G}$ is $f'_!$-acyclic as required. $\square$

Theorem 3.7. *Suppose $f : X \to Y$ is a continuous map of locally compact spaces. If $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a soft sheaf, then $f_! \mathcal{F}$ is soft.*

Proof. Let $K \subset Y$ be compact. Consider the pullback square

$$\begin{tikzcd} f^{-1}(K) \arrow[r, hook] \arrow[d, "f'"] & X \arrow[d, "f"] \\ K \arrow[r, hook] & Y, \end{tikzcd}$$

where f' is the restriction of f to $f^{-1}(K)$. By Theorem 3.6, we have an isomorphism

$$(f_! \mathcal{F})|_K \cong (f'_!)(\mathcal{F}|_{f^{-1}(K)}).$$

Applying $\Gamma(K, -)$ then yields an isomorphism $\Gamma(K, f_! \mathcal{F}) \cong \Gamma_c(f^{-1}(K), \mathcal{F})$. Now by Proposition 2.5, the restriction map $\Gamma_c(X, \mathcal{F}) \to \Gamma_c(f^{-1}(K), \mathcal{F})$ is surjective. But $\Gamma_c(X, \mathcal{F}) \subset \Gamma(Y, f_! \mathcal{F})$, so this shows that $\Gamma(Y, f_! \mathcal{F}) \to \Gamma(K, f_! \mathcal{F})$ is surjective, and hence that $f_! \mathcal{F}$ is soft. $\square$

Corollary 3.8. *Suppose X, Y and Z are locally compact spaces, and $f : X \to Y$ and $g : Y \to Z$ are continuous maps. We have a canonical natural isomorphism $R(g \circ f)_! \cong Rg_! \circ Rf_!$.*

Proof. This follows from Theorem 3.7 and the theorem on the composition of total derived functors. □

We conclude this section by discussing two special long exact sequences. The first is known as a *Mayer-Vietoris sequence*:

Proposition 3.9. *Suppose $A, B \subset X$ are closed subsets of X. Then for any sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$, we have a long exact sequence of the following form:*

$$\begin{tikzcd}
0 \arrow[r] & \Gamma_c(A \cup B, \mathcal{F}) \arrow[r] & \Gamma_c(A, \mathcal{F}) \oplus \Gamma_c(B, \mathcal{F}) \arrow[r] & \Gamma_c(A \cap B, \mathcal{F}) \arrow[dll, out=-30, in=150] \\
 & H^1_c(A \cup B, \mathcal{F}) \arrow[r] & H^1_c(A, \mathcal{F}) \oplus H^1_c(B, \mathcal{F}) \arrow[r] & H^1_c(A \cap B, \mathcal{F}) \arrow[dll, out=-30, in=150] \\
 & H^2_c(A \cup B, \mathcal{F}) \arrow[r] & H^2_c(A, \mathcal{F}) \oplus H^2_c(B, \mathcal{F}) \arrow[r] & \cdots
\end{tikzcd}$$

Proof. This is the long exact sequence associated to the short exact sequence (2.1) and the functor $\Gamma_c(X, -)$. □

We refer to the second as the *open/closed long exact sequence*:

Proposition 3.10. *Let $Z \subset X$ be a closed subset of X, and $U := X \setminus Z$ its open complement. Then for any sheaf $\mathcal{F} \in \mathbf{Sh}(X, k)$, we have a long exact sequence of the form:*

$$\begin{tikzcd}
0 \arrow[r] & \Gamma_c(U, \mathcal{F}) \arrow[r] & \Gamma_c(X, \mathcal{F}) \arrow[r] & \Gamma_c(Z, \mathcal{F}) \arrow[dll, out=-30, in=150] \\
 & H^1_c(U, \mathcal{F}) \arrow[r] & H^1_c(X, \mathcal{F}) \arrow[r] & H^1_c(Z, \mathcal{F}) \arrow[dll, out=-30, in=150] \\
 & H^2_c(U, \mathcal{F}) \arrow[r] & H^2_c(X, \mathcal{F}) \arrow[r] & \cdots
\end{tikzcd}$$

Proof. Let $i : Z \to X$ and $j : U \to X$ denote the inclusions, and consider the following sequence of sheaves on X:

$$0 \longrightarrow j_! j^* \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_* i^* \mathcal{F} \longrightarrow 0; \tag{3.4}$$

by (1.1), this sequence is exact. Now the functor $\Gamma_c(U, -)$ can be expressed as a composition

$$\Gamma_c(U, -) \cong \Gamma_c(X, j_!(-)).$$

By Corollary 3.8, we have

$$R\Gamma_c(U, -) \cong R\Gamma_c(X, j_!(-));$$

note that $Rj_! = j_!$ since $j_!$ is exact. Using this identification with the long exact sequence associated to (3.4) and $\Gamma_c(X,-)$ produces the desired long exact sequence. □

We can use the open/closed long exact sequence to give the following characterisation of softness:

Corollary 3.11. *A sheaf $\mathcal{F} \in \mathbf{Sh}(X,k)$ is soft if and only if $H^1_c(U,\mathcal{F}|_U) = 0$ for every open subset $U \subset X$.*

Proof. If $\mathcal{F}$ is soft, then $\mathcal{F}|_U$ is also soft, and it follows that $H^1_c(U,\mathcal{F}|_U) = 0$ since a soft sheaf is $\Gamma_c(U,-)$ acyclic by Theorem 2.9.

Conversely, assume $H^1_c(U,\mathcal{F}|_U) = 0$ for every open subset $U \subset X$. We will prove that $\mathcal{F}$ is soft using the characterisation of Proposition 2.5. Let $Z \subset X$ be a closed subset; we wish to show that the restriction map $\Gamma_c(X,\mathcal{F}) \to \Gamma_c(Z,\mathcal{F})$ is surjective. But with $U = X \setminus Z$, a piece of the long exact cohomology sequence of Proposition 3.10 is

$$\Gamma_c(X,\mathcal{F}) \longrightarrow \Gamma_c(Z,\mathcal{F}) \longrightarrow H^1_c(U,\mathcal{F}) \longrightarrow H^1_c(X,\mathcal{F}),$$

and $H^1_c(U,\mathcal{F}) = H^1_c(X,\mathcal{F}) = 0$ by assumption. □

4. Computation of $H^i_c(\mathbb{R}^n)$

In this section, we compute the compactly supported cohomology of $\mathbb{R}^n$. To build some expectation for the answer, consider the special case of cohomology with coefficients in the field $k = \mathbb{R}$. Since the de Rham complex is a soft resolution of the constant sheaf $\mathbb{R}$, it can be used to compute both compactly supported and non-compactly supported cohomology. Poincaré duality says that we have a perfect pairing:

$$H^i(\mathbb{R}^n,\mathbb{R}) \times H^{n-i}_c(\mathbb{R}^n,\mathbb{R}) \longrightarrow H^n_c(\mathbb{R}^n,\mathbb{R}) \xrightarrow{\int} \mathbb{R},$$

where the first map is given by the wedge product, and the second is an isomorphism given by integration. Since $\mathbb{R}^n$ is contractible,

$$H^i(\mathbb{R}^n,\mathbb{R}) = \begin{cases} \mathbb{R} & \text{if } i = 0 \\ 0 & \text{if } i > 0, \end{cases}$$

so we have

$$H^i_c(\mathbb{R}^n,\mathbb{R}) = \begin{cases} 0 & \text{if } 0 \le i < n \\ \mathbb{R} & \text{if } i = n. \end{cases}$$

We will see that we get the essentially same answer if we consider compactly supported cohomology with coefficients in an arbitrary ring k.

Fix a smooth manifold X, with or without boundary. Primarily, we are interested in the cases $X = \mathbb{R}^n$ and $X = [0,1]$. We write $\mathcal{C}_X$ for the sheaf of continuous functions on X, and $\mathcal{C}^\infty_X$ for the sheaf of smooth functions on X. In fact, these sheaves are both soft. This will follow from the existence of smooth bump functions:

Lemma 4.1. *Given a closed subset $Z \subset X$ and an open subset $U \subset X$ containing Z, there exists a smooth function $\psi : X \to \mathbb{R}$ such that $0 \le \psi \le 1$ on X, $\psi|_Z$ is the constant function $1 : Z \to \mathbb{R}$, and $\mathrm{Supp}(\psi) \subset U$.*

Proof. See [Lee, Proposition 2.25]. □

Proposition 4.2. *Let X be a smooth manifold with or without boundary. Any sheaf $\mathcal{F}$ of $\mathcal{C}_X^\infty$-modules on X is soft.*

Proof. Suppose $K \subset X$ is compact, and $s \in \Gamma(K, \mathcal{F})$. By Proposition 1.2, s is the restriction of a section $\tilde{s} \in \Gamma(U, \mathcal{F})$, where $U \subset X$ is an open superset of K. Choose a compact set $L \subset U$ that contains K in its interior. By Lemma 4.1, there exists a function $\psi : X \to \mathbb{R}$ such that $\psi|_K = 1$ and $\mathrm{Supp}(\psi) \subset L^\circ$. The section $\psi\tilde{s} \in \Gamma(U, \mathcal{F})$ restricts to s on K, and since it is zero on $U \setminus L$, it can be extended by zero to a global section of $\mathcal{F}$. So $\mathcal{F}$ is soft, as required. □

In particular, $\mathcal{C}_X$ is a soft sheaf, since the homomorphism $\mathcal{C}_X^\infty \to \mathcal{C}_X$ of sheaves of rings gives $\mathcal{C}_X$ the structure of a $\mathcal{C}_X^\infty$-module.

Now that these preliminary results have been established, we can begin our computation of $H_c^i(\mathbb{R}^n, k)$. As a first step, we prove that the unit interval $X = [0, 1]$ is acyclic, in the following sense:

Definition 4.3. A compact topological space X is said to be *acyclic* if

$$H^i(X, \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } i = 0 \\ 0 & \text{if } i > 0. \end{cases}$$

Lemma 4.4. *Suppose X is a compact, acyclic topological space, and A is an abelian group. If A_X denotes the constant sheaf on X of stalk A, then we have:*

$$H^i(X, A_X) = \begin{cases} A & \textit{if } i = 0 \\ 0 & \textit{if } i > 0. \end{cases}$$

Proof. Firstly, note that since $H^0(X, \mathbb{Z}_X) = \mathbb{Z}$, X is connected. Hence any locally constant function $X \to A$ is constant, which implies $H^0(X, A_X) = A$.

It remains to show that $H^i(X, A_X) = 0$ for all $i > 0$. Since A is the direct limit of its finitely generated subgroups, we can use Theorem 3.3 reduce to the case where A is finitely generated. Assuming this, choose a finitely generated free resolution of A_X:

$$0 \longrightarrow \mathbb{Z}_X^{\oplus m} \longrightarrow \mathbb{Z}_X^{\oplus n} \longrightarrow A_X \longrightarrow 0;$$

this is possible since as a PID, $\mathbb{Z}$ has global dimension 1. From the long exact cohomology sequence, it follows easily that $H^i(X, A_X) = 0$ for all $i > 0$. □

Let $S^1 \subset \mathbb{C}$ denote the circle group. Recall that the exponential map

$$\varepsilon : \mathbb{R} \longrightarrow S^1, \qquad \varepsilon(x) = e^{2\pi i x},$$

is a covering map; in particular, it enjoys the following lifting property: for every interval $J \subset \mathbb{R}$ and every continuous map $f : J \to S^1$, there exists a continuous map $g : J \to \mathbb{R}$ making the following diagram commute:

$$\begin{tikzcd} & \mathbb{R} \arrow[d, "\varepsilon"] \\ J \arrow[ur, "g"] \arrow[r, "f"'] & S^1. \end{tikzcd} \tag{4.1}$$

Now consider the sheaf $\mathcal{C}_X(S^1)$ locally defined continuous maps $X \to S^1$. From the lifting property (4.1), it follows that the sequence of sheaves of abelian groups

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{C}_X \xrightarrow{\varepsilon \circ (-)} \mathcal{C}_X(S^1) \longrightarrow 1 \tag{4.2}$$

is exact; (4.2) is called the *exponential exact sequence.*

Lemma 4.5. *The sheaf* $\mathcal{C}_X(S^1)$ *on* $X = [0,1]$ *is soft.*

Proof. Let $V \subset X$ be an open subset, and $f \in \Gamma(V, \mathcal{C}_X(S^1))$ a section over V. Write V as a disjoint union of relatively open intervals: $V = \cup_i V_i$, $i \in I$. For each $i \in I$, write $f_i := f|_{V_i}$, and use the lifting property (4.1) to lift each f_i to a map $g_i : V_i \to \mathbb{R}$ such that $f_i = \varepsilon \circ g_i$. Let $g : V \to \mathbb{R}$ be the induced map. Now since $\mathcal{C}_X$ is soft, g extends to a global function $\tilde{g} : X \to \mathbb{R}$. If we set $\tilde{f} := \varepsilon \circ \tilde{g} \in \Gamma(X, \mathcal{C}_X(S^1))$, then we have $\tilde{f}|_V = f$, which shows that $\mathcal{C}_X(S^1)$ is flasque, and hence soft. □

Proposition 4.6. *The unit interval* $X = [0,1]$ *is acyclic.*

Proof. It follows from Lemma 4.5 that the exponential exact sequence (4.2) is a soft resolution of the constant sheaf $\mathbb{Z}_X$. If we apply $\Gamma(X, -) = \Gamma_c(X, -)$, we see immediately that $H^0(X, \mathbb{Z}) = \mathbb{Z}$ and $H^i(X, \mathbb{Z}) = 0$ for all $i \geq 2$, while the fact that $H^1(X, \mathbb{Z}) = 0$ follows from another application of (4.1). □

To generalise this result to higher dimensions, we need the following result, which is known as the Vietoris-Begle mapping theorem:

Theorem 4.7. *Suppose* $f : X \to Y$ *is proper, with acyclic fibres. Then the unit natural transformation* $\eta : \mathrm{id} \to f_* \circ f^*$ *is a natural isomorphism, and it induces an isomorphism of functors* $\mathrm{D}^+(Y, k) \to \mathrm{D}^+(k)$*:*

$$R\Gamma_c(Y, -) \xrightarrow{\sim} R\Gamma_c(X, f^*(-)).$$

In particular, for all sheaves $\mathcal{G} \in \mathbf{Sh}(Y, k)$ *and all* $i \in \mathbb{Z}$*, the inverse image induces an isomorphism on cohomology:*

$$f^* : H^i_c(Y, \mathcal{G}) \xrightarrow{\sim} H^i_c(X, f^*\mathcal{G}).$$

Proof. Let $\mathcal{G} \in \mathbf{Sh}(Y, k)$. We claim that $(\eta_{\mathcal{G}})_y : \mathcal{G}_y \to (f_* f^* \mathcal{G})_y$ is an isomorphism for all $y \in Y$. If we pull $\mathcal{G}$ back to X_y along the fibre diagram

$$\begin{tikzcd} X_y \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \{y\} \arrow[r] & Y, \end{tikzcd}$$

we see that $(f^*\mathcal{G})|_{X_y}$ is the constant sheaf of stalk $\mathcal{G}_y$ on X_y. Let $V \subset Y$ be an open neighbourhood of y, and consider the diagram

$$\begin{tikzcd} \mathcal{G}_y \arrow[r, "(\eta_{\mathcal{G}})_y"] & (f_* f^* \mathcal{G})_y \arrow[r, "\sim"] & \Gamma(X_y, (\mathcal{G}_y)_{X_y}) \\ \mathcal{G}(V) \arrow[u] \arrow[r, "\eta_{\mathcal{G}}(V)"] & (f^*\mathcal{G})(f^{-1}(V)), \arrow[u] \arrow[ur, "\mathrm{res}"'] & \end{tikzcd}$$

where the top right arrow is the isomorphism of Lemma 2.6. If $s \in \mathcal{G}(V)$ is a section, then the restriction of the pullback $\eta_{\mathcal{G}}(V)(s) = f^* s$ to X_y satisfies

$$(f^* s)_x = s_y = (f^* s)_{x'}$$

in $\mathcal{G}_y$ for all points $x, x' \in X_y$. So the given map $\mathcal{G}_y \to \Gamma(X_y, (\mathcal{G}_y)_{X_y})$ is the canonical one, which sends an element of $\mathcal{G}_y$ to the corresponding constant function on X_y.

But this map is an isomorphism by Proposition 4.4, and the fact that the fibre X_y is acyclic. Therefore, the unit transformation η is a natural isomorphism.

By postcomposing η with $\Gamma_c(Y, -)$, we get an isomorphism of functors

$$\Gamma_c(Y, -) \xrightarrow{\sim} \Gamma_c(Y, f_* \circ f^*(-)). \tag{4.3}$$

At this point, we wish to use the theorem of the composition of total derived functors to compute the derived functor on the right hand side of (4.3). Since f_* sends injective sheaves to injective sheaves, what we have to check is that $f^*\mathcal{G}$ is f_*-acyclic for all sheaves $\mathcal{G} \in \mathbf{Sh}(Y, k)$. But by Proposition 3.5, Proposition 4.4, and the fact that X_y is an acyclic space, we have

$$(R^i f_*(f^*\mathcal{G}))_y \cong H^i(X_y, (f^*\mathcal{G})|_{X_y}) = H^i(X_y, (\mathcal{G}_y)|_{X_y}) = 0$$

for all $y \in Y$ and $i > 0$. So the derived functor of the right hand side of (4.3) can be written as a composition

$$R\Gamma_c(Y, Rf_* \circ f^*(-)) \cong R\Gamma_c(X, f^*(-));$$

note that $R\Gamma_c(Y, Rf_*(-)) \cong R\Gamma_c(X, -)$ since f_* is proper. Therefore, by deriving (4.3), we obtain the desired natural isomorphism of derived functors. □

Corollary 4.8. *The n-cube $C^n := [0, 1]^n$ is acyclic.*

Proof. We prove this by induction on n, with the base case $n = 1$ being Proposition 4.6. Assume C^n is acyclic for some $n \geq 1$, and consider the projection map

$$p : C^{n+1} \longrightarrow C^n \ , \qquad (x_1, \dots, x_n, x_{n+1}) \mapsto (x_1, \dots, x_n).$$

Since p is proper with acyclic fibres, we can apply Theorem 4.7 with $\mathcal{G}$ equal to the constant sheaf $\mathbb{Z}$ on C^n to conclude that

$$H^i(C^{n+1}, \mathbb{Z}) \cong H^i(C^n, \mathbb{Z})$$

for all $i \geq 0$. This shows C^{n+1} is acyclic, and closes the induction. □

The next step is to compute the cohomology of the n-sphere

$$S^n := \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}.$$

Proposition 4.9. *The cohomology of the n-sphere (for $n \geq 1$) is*

$$H^i(S^n, k) = \begin{cases} k & \text{if } i = 0 \text{ or } i = n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. First of all, note that $H^0(S^n, k) = k$ since S^n is connected, so we only have to compute the higher cohomology groups. We will do this using induction on $n \geq 1$ and the Mayer-Vietoris sequence. To set this up, note that S^n is the union of two hemispheres:

$$S^n_+ := \{(x_1, x_2, \dots, x_{n+1}) \in S^n \mid x_{n+1} \geq 0\}$$

and

$$S^n_- := \{(x_1, x_2, \dots, x_{n+1}) \in S^n \mid x_{n+1} \leq 0\}.$$

Moreover, both S^n_+ and S^n_- are acyclic, since they are homeomorphic to the n-cube C^n via projection onto the first n coordinates. Also, the intersection $S^n_+ \cap S^n$ is homeomorphic to S^{n-1}.

We now proceed with the proof. Since the semicircles S^1_+ and S^1_- are acyclic and $S^1_+ \cap S^1_-$ is the union of two points, the Mayer-Vietoris sequence shows immediately that $H^i(S^1, k) = 0$ for all $i \geq 2$. To compute $H^1(S^1, k) = 0$, the relevant piece of the sequence is

$$\Gamma(S^1_+, k) \oplus \Gamma(S^1_-, k) \longrightarrow \Gamma(S^1_+ \cap S^1_-, k) \longrightarrow H^1(S^1, k) \longrightarrow 0.$$

Since the image of the first map in $\Gamma(S^1_+ \cap S^1_-, k) = k \oplus k$ is generated by $(1, -1)$, we have

$$H^1(S^1, k) \cong (k \oplus k)/k(1, -1) \cong k.$$

Now assume the result holds for some $n \geq 1$. Using the Mayer-Vietoris sequence and the fact that S^n_+ and S^n_- are acyclic, we find that

$$H^i(S^{n+1}, k) \cong H^{i-1}(S^{n+1}_+ \cap S^{n+1}_-, k) \cong H^{i-1}(S^n, k)$$

for all $i \geq 2$. It remains to compute $H^1(S^{n+1}, k)$. Again, the relevant segment of the Mayer-Vietoris sequence is

$$\Gamma(S^{n+1}_+, k) \oplus \Gamma(S^{n+1}_-, k) \longrightarrow \Gamma(S^{n+1}_+ \cap S^{n+1}_-, k) \longrightarrow H^1(S^{n+1}, k) \longrightarrow 0.$$

This time, however, the first map is surjective, since $S^{n+1}_+ \cap S^{n+1}_- \cong S^n$ is connected. Therefore, $H^1(S^{n+1}, k) = 0$, and this completes the proof. $\square$

Theorem 4.10. *The compactly supported cohomology of $\mathbb{R}^n$ is*

$$H^i_{\mathrm{c}}(\mathbb{R}^n, k) = \begin{cases} k & \text{if } i = n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. First of all, note that $H^0_{\mathrm{c}}(\mathbb{R}^n, k) = 0$, since a constant function on $\mathbb{R}^n$ whose support is compact must be zero. Now recall that $\mathbb{R}^n$ is homeomorphic to $S^n \setminus \{N\}$ via stereographic projection, where $N = (0, 0, \ldots, 0, 1) \in S^n$ is the 'north pole'. The open/closed long exact sequence of Proposition 3.10 associated to N shows that

$$H^i_{\mathrm{c}}(\mathbb{R}^n, k) \cong H^i(S^n, k) \tag{4.4}$$

for all $i \geq 2$, while the beginning of the sequence reads

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(S^n, k) \arrow[r] & \Gamma(\{N\}, k) \arrow[dll] \\ H^1_{\mathrm{c}}(\mathbb{R}^n, k) \arrow[r] & H^1(S^n, k) \arrow[r] & 0. \end{tikzcd}$$

But the map $\Gamma(S^n, k) \to \Gamma(\{N\}, k)$ is an isomorphism, so (4.4) is an isomorphism also for $n = 1$. The computation of $H^i(S^n, k)$ in Proposition 4.9 completes the proof. $\square$

5. Cohomological dimension

In this section, we introduce a notion of dimension for locally compact spaces, which is based on homological algebra.

Definition 5.1. Suppose X is a locally compact space. We say that X has *finite cohomological dimension* if there exists an integer n such that for all integers $k > n$ and all sheaves $\mathcal{F} \in \mathbf{Sh}(X)$, we have $H^k_{\mathrm{c}}(X, \mathcal{F}) = 0$. The *cohomological dimension* of X, denoted $\dim_{\mathrm{c}}(X)$, is defined to be the smallest integer with this property.

Our main goal is to show that if X is an n-dimensional topological manifold, then $\dim_c(X) = n$. There are two steps to this: we first compute that $\dim_c(\mathbb{R}^n) = n$, and then show dimension is, in an appropriate sense, a local property.

To compute $\dim_c(\mathbb{R}^n)$, we will use the following inequality:

Lemma 5.2. *Suppose X is a locally compact space with $\dim_c(X) < \infty$. Then*

$$\dim_c(X \times \mathbb{R}) \leq \dim_c(X) + 1.$$

Proof. Assume, for contradiction, that there exists an integer $m > \dim_c(X) + 1$, a sheaf $\mathcal{F} \in \mathbf{Sh}(X \times \mathbb{R})$, and a nonzero cohomology class $\gamma \in H^m_c(X \times \mathbb{R}, \mathcal{F})$. Let $\mathcal{Z}$ be the collection of closed subsets of $X \times \mathbb{R}$ of the form $X \times F$, where $F \subset \mathbb{R}$ is closed. By Lemma 5.3 below, there exists a minimal $Z = X \times F \in \mathcal{Z}$ such that $\gamma|_Z \neq 0$. If F consists of three or more points, there exists point $t \in F$ such that the sets

$$Z_+ := X \times (F \cap (-\infty, t]), \quad Z_- := X \times (F \cap [t, \infty))$$

are proper subsets of Z. Note that $Z_+ \cup Z_- = Z$ and $Z_+ \cap Z_- = X \times \{t\}$, and consider the following piece of the corresponding Mayer-Vietoris sequence (Proposition 3.9):

$$H^{m-1}_c(X \times \{t\}, \mathcal{F}) \longrightarrow H^m_c(Z, \mathcal{F}) \longrightarrow H^m_c(Z_+, \mathcal{F}) \oplus H^m_c(Z_-, \mathcal{F}).$$

Since $m - 1 > \dim_c(X)$ and $X \times \{t\} \cong X$, we have $H^{m-1}_c(X \times \{t\}, \mathcal{F}) = 0$. Hence the restriction map $H^m_c(Z, \mathcal{F}) \to H^m_c(Z_+, \mathcal{F}) \oplus H^m_c(Z_-, \mathcal{F})$ in injective, so one of $\gamma|_{Z_+}$ and $\gamma|_{Z_-}$ must be nonzero. But this contradicts the minimality of Z. The case where F consists of two points is treated similarly. If F is a single point then Z is homeomorphic to X, and we already have a contradiction. □

Lemma 5.3. *Let $\mathcal{Z}$ be a family of closed subsets of X that is stable under arbitrary intersections (so in particular, $X \in \mathcal{Z}$). Suppose $\mathcal{F} \in \mathbf{Sh}(X, k)$ is a sheaf, and $\gamma \in H^i_c(X, \mathcal{F})$ is a nonzero cohomology class. Then the ordered set*

$$\{Z \in \mathcal{Z} \mid \ \gamma|_Z \neq 0 \text{ in } H^i_c(Z, \mathcal{F}|_Z)\}$$

contains minimal elements.

Proof. This follows immediately from Zorn's lemma and Corollary 3.4. □

Proposition 5.4. *The cohomological dimension of $\mathbb{R}^n$ is n.*

Proof. The calculation of the compactly supported cohomology of $\mathbb{R}^n$ (Theorem 4.10) shows that $\dim_c(\mathbb{R}^n) \geq n$. On the other hand, since the cohomological dimension of a point is 0, Lemma 5.2 combined with induction shows that $\dim_c(\mathbb{R}^n) \leq n$. □

The notion 'dimension at a point' is captured by the following definition:

Definition 5.5. We define the *local cohomological dimension of X at $x \in X$* to be the integer

$$(\dim_c)_x(X) := \inf\{\dim_c(U) \mid U \subset X \text{ is an open neighbourhood of } x\}.$$

Proving that 'cohomological dimension is local' amounts to relating the dimension of X to the dimension at each of its points. If $x \in X$, then it is immediate from the definition that $(\dim_c)_x(X) \leq \dim_c(X)$, so the natural thing to try to prove is that equality is achieved for some x.

Proposition 5.6. *If W is a locally closed subspace of X, then $\dim_c(W) \leq \dim_c(X)$.*

Proof. Writing $j : W \hookrightarrow X$ for the inclusion, this follows immediately from the identity

$$R\Gamma_c(W, -) \cong R\Gamma_c(X, j_!(-)),$$

which we saw in the course of the proof of Proposition 3.10. □

The following technical lemma is the key to proving that cohomological dimension is local, and will also be crucial in our discussion of Verdier duality in the next section.

Lemma 5.7. *Suppose* $n \geq \dim_c(X)$*, and we have an exact sequence in* $\mathbf{Sh}(X, k)$*:*

$$\mathcal{F}^0 \longrightarrow \mathcal{F}^1 \longrightarrow \cdots \longrightarrow \mathcal{F}^{n-1} \longrightarrow \mathcal{F}^n \longrightarrow 0.$$

If $\mathcal{F}^i$ *is soft for all* $0 \leq i \leq n-1$*, then* $\mathcal{F}^n$ *is also soft. In particular, any sheaf* $\mathcal{F} \in \mathbf{Sh}(X, k)$ *that admits a soft resolution from the left is itself soft.*

Proof. By Corollary 3.11 and Lemma 2.2, it suffices to show $H^1_c(X, \mathcal{F}^n) = 0$. For each $0 \leq i \leq n-1$, let $\mathcal{K}^i := \mathrm{Ker}(\mathcal{F}^i \to \mathcal{F}^{i+1})$. We then have short exact sequences

$$0 \longrightarrow \mathcal{K}^{n-1} \longrightarrow \mathcal{F}^{n-1} \longrightarrow \mathcal{F}^n \longrightarrow 0$$

and

$$0 \longrightarrow \mathcal{K}^i \longrightarrow \mathcal{F}^i \longrightarrow \mathcal{K}^{i+1} \longrightarrow 0$$

for $0 \leq i \leq n-2$. Since the $\mathcal{F}^i$ are soft for $i < n$, the corresponding long exact cohomology sequences show that

$$H^1_c(X, \mathcal{F}^n) \cong H^2_c(X, \mathcal{K}^{n-1}) \cong \cdots \cong H^{n+1}_c(X, \mathcal{K}^0).$$

But since $n+1 > \dim_c(X)$, this shows $H^1_c(X, \mathcal{F}^n) = 0$, as required. □

Theorem 5.8. *Cohomological dimension is a local property, in the sense of the following formula:*

$$\dim_c(X) = \sup\{(\dim_c)_x(X) \mid x \in X\}.$$

Proof. Let $m = \sup\{(\dim_c)_x(X) \mid x \in X\}$. If $x \in X$ is a point, then the definition of $(\dim_c)_x(X)$ immediately implies $(\dim_c)_x(X) \leq \dim_c(X)$, so $m \leq \dim_c(X)$. To prove $m \geq \dim_c(X)$, we must show that $H^k_c(X, \mathcal{G}) = 0$ for all sheaves $\mathcal{G} \in \mathbf{Sh}(X)$ and integers $k > m$. Consider a resolution of $\mathcal{G}$:

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{F}^0 \longrightarrow \cdots \longrightarrow \mathcal{F}^{k-2} \longrightarrow \mathcal{F}^{k-1} \longrightarrow 0, \tag{5.1}$$

where the terms $\mathcal{F}^0, \dots, \mathcal{F}^{k-2}$ are all soft. Choose an open neighbourhood U_x around each point $x \in X$ such that $\dim_c(U_x) = (\dim_c)_x(X)$. Since

$$\dim_c(U_x) \leq m \leq k-1,$$

we can apply Lemma 5.7 to the restriction of (5.1) to U_x to conclude that $(\mathcal{F}^{k-1})|_{U_x}$ is soft. Since softness is a local property (this is not difficult to check), it follows that $\mathcal{F}^{k-1}$ is soft. Therefore, (5.1) is a soft resolution of $\mathcal{G}$. Applying $\Gamma_c(X, -)$ and taking cohomology in degree k, we find that $H^k_c(X, \mathcal{G}) = 0$, as required. □

Corollary 5.9. *If* X *is an* n*-dimensional topological manifold, then* $\dim_c(X) = n$*.*

Proof. We will show that $(\dim_c)_x(X) = n$ for each point $x \in X$, and then apply Theorem 5.8. Since x has a neighbourhood U which is homeomorphic to $\mathbb{R}^n$, we have $(\dim_c)_x(X) \leq n$ by definition of $(\dim_c)_x(X)$. On the other hand, if V is an arbitrary open neighbourhood of x, then we can find an open ball B such that $x \in B \subset V$. By Proposition 5.6, $\dim_c(V) \geq \dim_c(B) = n$, since $B \cong \mathbb{R}^n$. So $(\dim_c)_x(X) = n$, as claimed. □

6. Verdier duality

The subject of Verdier duality is motivated by the following question: if $f : X \to Y$ is a map of locally compact spaces, does $f_!$ necessarily have a right adjoint? If a right adjoint does exist, it will be denoted $f^!$. We have already seen in Proposition 1.6 that if $j : U \hookrightarrow Y$ is the inclusion of an open subset, then the restriction j^* is right adjoint to $j_!$, so $j^! = j^*$ in this case.

If $f^!$ exists, then the adjunction formula

$$\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!\mathcal{F}, \mathcal{G}) = \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}, f^!\mathcal{G})$$

implies that for each $\mathcal{G} \in \mathbf{Sh}(Y,k)$, the functor $\mathrm{Hom}(f_!(-), \mathcal{G})$ is represented by the sheaf $f^!\mathcal{G}$. So we are led to study the representability of contravariant functors

$$F : \mathbf{Sh}(X,k)^{\mathrm{op}} \to k{-}\mathbf{Mod}.$$

Theorem 6.1. *A contravariant functor $F : \mathbf{Sh}(X,k)^{\mathrm{op}} \to k{-}\mathbf{Mod}$ is representable if and only if it sends colimits in $\mathbf{Sh}(X,k)$ to limits in $k{-}\mathbf{Mod}$.*

Proof. If $\mathcal{F} \in \mathbf{Sh}(X,k)$, the functor $\mathrm{Hom}(-, \mathcal{F})$ sends colimits to limits — this is true in any category.

So suppose F sends colimits to limits. Given an open subset $U \subset X$, recall that k_U is our notation for the extension by zero $j_!(k)$ of the constant sheaf k on U (here $j : U \hookrightarrow X$ denotes the inclusion). We define a presheaf $\mathcal{F}$ on X by setting

$$\mathcal{F}(U) := F(k_U).$$

To define the restriction maps, suppose $V \subset U$ is a smaller open subset, and let $i : V \hookrightarrow U$ be the inclusion. Then the counit of the adjunction of Proposition 1.6 gives rise to a natural transformation

$$(j \circ i)_! \circ (j \circ i)^* = j_! \circ i_! \circ i^* \circ j^* \longrightarrow j_! \circ j^*,$$

whose component over the constant sheaf k on X is a map $\alpha_{V,U} : k_V \to k_U$. Applying F to $\alpha_{V,U}$ defines the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$.

Next, we wish to show that $\mathcal{F}$ is actually a sheaf. Given an open cover $\{U_i\}_{i\in I}$ of an open subset $U \subset X$, we have a canonical exact sequence in $\mathbf{Sh}(X,k)$:

$$\bigoplus\nolimits_{i,j} k_{U_i\cap U_j} \longrightarrow \bigoplus\nolimits_i k_{U_i} \longrightarrow k_U \longrightarrow 0.$$

Here the maps $k_{U_i\cap U_j} \to \oplus_i k_{U_i}$ are given by $\alpha_{U_i\cap U_j, U_i} - \alpha_{U_i \cap U_j, U_j}$, and exactness follows from (1.1). Since F sends colimits in $\mathbf{Sh}(X,k)$ to limits in $k{-}\mathbf{Mod}$, applying F produces a sequence of k-modules

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod\nolimits_i \mathcal{F}(U_i) \longrightarrow \prod\nolimits_{i,j} \mathcal{F}(U_i \cap U_j),$$

whose exactness says precisely that $\mathcal{F}$ is sheaf.

To prove that $\mathcal{F}$ is a representing object for F, we will establish first that every sheaf $\mathcal{G} \in \mathbf{Sh}(X,k)$ is a colimit of sheaves of the form k_U. We define an index category $\mathbf{J}_\mathcal{G}$ with objects

$$\mathrm{Ob}\,\mathbf{J}_\mathcal{G} := \{(U,s) \mid U \subset X \text{ is open}, \ s \in \mathcal{G}(U)\}$$

and morphisms

$$\mathrm{Hom}_{\mathbf{J}_\mathcal{G}}((U,s),(U',s')) = \begin{cases} \{*\} & \text{if } U \subset U' \text{ and } s'|_U = s \\ \varnothing & \text{otherwise,} \end{cases}$$

with composition being defined in the obvious way. We then define a diagram

$$D : \mathbf{J}_{\mathcal{G}} \to \mathbf{Sh}(X, k)$$

on an object $j = (U, s)$ by $D_j := k_U$, and on a morphism $(U, s) \to (U', s')$ by the canonical morphism $\alpha_{U,U'} : k_U \to k_{U'}$.

Now that the diagram D has been defined, it remains to show that $\mathcal{G}$ is a cocone from D, and in fact that $\mathcal{G}$ is the colimit of D. If $j = (U, s) \in \mathbf{J}_{\mathcal{G}}$, then since $\mathcal{G}(U) \cong \mathrm{Hom}_{\mathbf{Sh}(X,k)}(k_U, \mathcal{G})$ as k-modules, the section s defines a morphism $D_j \to \mathcal{G}$, and if $(U, s) \to (U', s')$ is a morphism in $\mathbf{J}_{\mathcal{G}}$, then the condition that $s'|_V = s$ implies the diagram

$$\begin{tikzcd} k_U \arrow[rr] \arrow[dr, "s"'] & & k_{U'} \arrow[dl, "s'"] \\ & \mathcal{G} & \end{tikzcd}$$

commutes. This shows $\mathcal{G}$ is a cocone from D. Suppose $\{t_j : D_j \to \mathcal{H} \mid j \in \mathbf{J}_{\mathcal{G}}\}$ is another cocone from D. If we define a morphism of sheaves $\mathcal{G} \to \mathcal{H}$ by sending a local section $s \in \mathcal{G}(U)$ to the section of $\mathcal{H}(U)$ corresponding to t_j, where $j = (U, s)$, then this is clearly the unique morphism $\mathcal{G} \to \mathcal{H}$ making the relevant triangles commute. Hence, $\mathcal{G}$ is the colimit of D.

Now because F sends colimits to limits, $F(\mathcal{G})$ is the limit of the diagram $F(D)$ in $k-\mathbf{Mod}$. But $\mathrm{Hom}(-, \mathcal{F})$ sends colimits to limits too, so we have

$$\begin{aligned} \mathrm{Hom}(\mathcal{G}, \mathcal{F}) &= \mathrm{Hom}(\varinjlim_j D_j, \mathcal{F}) \\ &\cong \varprojlim_j \mathrm{Hom}(D_j, \mathcal{F}) \\ &\cong \varprojlim_j F(D_j), \end{aligned}$$

since if $j = (U, s)$, then $\mathrm{Hom}(D_j, \mathcal{F}) = \mathrm{Hom}(k_U, \mathcal{F}) = \mathcal{F}(U) = F(D_j)$.

So for each $\mathcal{G} \in \mathbf{Sh}(X, k)$ we have an isomorphism $F(\mathcal{G}) \cong \mathrm{Hom}(\mathcal{G}, \mathcal{F})$. Naturality in $\mathcal{G}$ follows from the functoriality of all the constructions involved. □

Therefore, the functor $\mathrm{Hom}(f_!(-), \mathcal{G})$ is representable if it sends colimits to limits. Since $\mathrm{Hom}(-, \mathcal{G})$ sends colimits to limits, this will be true if $f_!$ preserves colimits. Now colimits are built out of coproducts and coequalizers, and since $\mathbf{Sh}(X, k)$ is an abelian category, coequalizers are essentially the same thing as cokernels. Since $f_!$ always preserves coproducts, we have proved the following result:

Corollary 6.2. *If $f : X \to Y$ is a map of locally compact spaces, then if $f_!$ is an exact functor, a right adjoint*

$$f^! : \mathbf{Sh}(Y, k) \to \mathbf{Sh}(X, k)$$

to $f_!$ exists.

If $j : U \hookrightarrow Y$ is the inclusion of an open subset, then extension by zero $j_!$ is exact, so Corollary 6.2 reproves the existence of $j^!$. There is another special case when the functor $f_!$ is exact: when f is the inclusion of a closed subset.

Proposition 6.3. *Let $i : Z \hookrightarrow Y$ be the inclusion of a closed subset Z into Y. Then $i_!$ has a right adjoint $i^!$.*

To generalise, we have to change our strategy. Idea: move to the derived category, where we can replace $\mathcal{F}$ by a soft sheaf (which is $f_!$-acyclic), use this to define a right adjoint $f^!$ to $Rf_!$. We define a soft replacement for the constant sheaf k. In fact, a truncation of the Godement resolution of k has the required properties.

Proposition 6.4. *Suppose* $\dim_c(X) = n$. *Then there exists a bounded resolution of the constant sheaf* k*:*

$$0 \longrightarrow k \longrightarrow \mathcal{K}^0 \longrightarrow \mathcal{K}^1 \longrightarrow \cdots \longrightarrow \mathcal{K}^n \longrightarrow 0,$$

whose terms $\mathcal{K}^i \in \mathbf{Sh}(X, k)$ *are soft and* k*-flat.*

Proof. Given $\mathcal{F} \in \mathbf{Sh}(X, k)$, we define a sheaf

$$G(\mathcal{F}) := \prod_{x \in X} (i_x)_* \mathcal{F}_x,$$

where $i_x : \{x\} \to X$ is the inclusion. Note that there is a canonical injective map $\mathcal{F} \hookrightarrow G(\mathcal{F})$. Moreover, $G(\mathcal{F})$ is flasque, and hence soft by Proposition 1.2. Now suppose $\mathcal{F}$ is k-flat; we claim that $G(\mathcal{F})$ is k-flat too. Note that the stalk of $G(\mathcal{F})$ at a point $x \in X$ is given by

$$G(\mathcal{F})_x = \varinjlim_{U \ni x} \prod_{x' \in U} \mathcal{F}_{x'}.$$

Since direct limits of flat k-modules are flat (a consequence of the fact that the tensor product commutes with all colimits), we need to know that an arbitrary direct product of flat k-modules is flat. This is true, because k was assumed to be Noetherian; see [Iv, VI.1.4]. Finally, observe that the map on stalks $\mathcal{F}_x \hookrightarrow G(\mathcal{F})_x$ is a split injection. It follows from this that $G(\mathcal{F})/\mathcal{F}$ is k-flat.

To define the resolution, let $\mathcal{K}^0 := G(k)$, so that there is an injective map $k \hookrightarrow \mathcal{K}^0$, which we label d^{-1}. Assuming a complex

$$0 \longrightarrow k \longrightarrow \mathcal{K}^0 \longrightarrow \cdots \longrightarrow \mathcal{K}^{i-1} \xrightarrow{d^{i-1}} \mathcal{K}^i$$

has already been defined for some $0 \le i \le n-2$, let $\mathcal{K}^{i+1} := G(\mathrm{Coker}(d^{i-1}))$, and let d^i be given by the composition $\mathcal{K}^i \twoheadrightarrow \mathrm{Coker}(d^{i-1}) \hookrightarrow \mathcal{K}^{i+1}$. Finally, define $\mathcal{K}^n$ to be the cokernel of $d^{n-2} : \mathcal{K}^{n-2} \to \mathcal{K}^{n-1}$, and let d^{n-1} be the canonical map.

Note that $\mathcal{K}^\cdot$ is a soft resolution of k: the terms $\mathcal{K}^i$ with $i < n$ are all soft by construction, while the softness of $\mathcal{K}^n$ follows from Lemma 5.7. It remains to show that the $\mathcal{K}^i$ are k-flat. Since the sheaf k is k-flat, $\mathcal{K}^0 = G(k)$ is k-flat, and so is $\mathrm{Coker}(d^{-1}) = G(k)/k$. Now assume for some $0 \le i \le n-2$ that both $\mathcal{K}^i$ and $\mathrm{Im}(d^i) \cong \mathrm{Coker}(d^{i-1})$ are k-flat. Then $\mathcal{K}^{i+1} = G(\mathrm{Coker}(d^{i-1}))$ is k-flat, and if we consider the short exact sequence

$$0 \longrightarrow \mathrm{Ker}(d^i) \longrightarrow \mathcal{K}^i \xrightarrow{d^i} \mathrm{Im}(d^i) \longrightarrow 0,$$

then since $\mathcal{K}^i \cong G(\mathrm{Im}(d^{i-1})) = G(\mathrm{Ker}(d^i))$, we see that $\mathrm{Im}(d^i)$ is k-flat as well. Finally, the fact that $\mathcal{K}^n$ is k-flat follows from essentially the same observation. $\square$

Here is how we will use Proposition 6.4:

Lemma 6.5. *Suppose* $\mathcal{F}^\cdot, \mathcal{L}^\cdot \in \mathbf{Com}^+(\mathbf{Sh}(X, k))$ *are bounded below complexes of sheaves, such that* $\mathcal{L}^\cdot$ *consists of* k*-flat objects. Then the natural map* $\mathcal{F}^\cdot \to \mathcal{F} \otimes_k \mathcal{L}^\cdot$ *is a quasi-isomorphism.*

Proof. Consider the spectral sequence of the double complex $(\mathcal{F}^p \otimes_k \mathcal{L}^q)_{p,q\in\mathbb{Z}}$ whose totalisation is $\mathcal{F} \otimes_k \mathcal{L}^\bullet$. □

Lemma 6.6. *Suppose X has finite cohomological dimension. If $\mathcal{F}, \mathcal{L} \in \mathbf{Sh}(X,k)$ such that $\mathcal{L}$ is soft and k-flat, then $\mathcal{F} \otimes_k \mathcal{L}$ is soft as well.*

Proof. We will show that $\mathcal{F} \otimes_k \mathcal{L}$ admits a soft resolution from the left, and then apply Lemma 5.7. Given an inclusion of an open subset $j : U \hookrightarrow X$ and a section $s \in \Gamma(U, \mathcal{F})$, there is a natural morphism of sheaves $k_U := j_!(k) \to \mathcal{F}$ which sends 1_U to s. Here k is the constant sheaf on U. Letting $\mathcal{F}^1 := \oplus_{(U,s)} k_U$ be the direct sum over all such pairs, we obtain a surjective map $\mathcal{F}^1 \twoheadrightarrow \mathcal{F}$. If $\mathcal{K}^1$ is the kernel of this map, we can in the same way find a surjection $\mathcal{F}^2 \twoheadrightarrow \mathcal{K}^1$. Continuing like this produces an infinite exact sequence

$$\cdots \longrightarrow \mathcal{F}^3 \longrightarrow \mathcal{F}^2 \longrightarrow \mathcal{F}^1 \longrightarrow \mathcal{F} \longrightarrow 0.$$

Since $\mathcal{L}$ is k-flat, the sequence

$$\cdots \longrightarrow \mathcal{F}^3 \otimes_k \mathcal{L} \longrightarrow \mathcal{F}^2 \otimes_k \mathcal{L} \longrightarrow \mathcal{F}^1 \otimes_k \mathcal{L} \longrightarrow \mathcal{F} \otimes_k \mathcal{L} \longrightarrow 0$$

remains exact. Moreover, each term $\mathcal{F}^i \otimes_k \mathcal{L}$ is soft, since it is a direct sum of terms of the form $k_U \otimes_k \mathcal{L} \cong \mathcal{L}_U$, and $\mathcal{L}_U = j_!(\mathcal{L}|_U)$ is soft by Theorem 3.7. Therefore, by Lemma 5.7, $\mathcal{F} \otimes_k \mathcal{L}$ is soft. □

Proposition 6.7. *Suppose X has finite cohomological dimension. Let $\mathcal{L}$ be a soft, k-flat sheaf in $\mathbf{Sh}(X,k)$. For each $\mathcal{G} \in \mathbf{Sh}(Y,k)$, the contravariant functor*

$$\mathrm{Hom}(f_!(- \otimes_k \mathcal{L}), \mathcal{G}) : \mathbf{Sh}(X,k)^{\mathrm{op}} \longrightarrow k\text{--}\mathbf{Mod}$$

is representable: there exists a sheaf $f^!(\mathcal{L},\mathcal{G}) \in \mathbf{Sh}(X,k)$ and a natural isomorphism

$$\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(- \otimes_k \mathcal{L}), \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{Sh}(X,k)}(-, f^!(\mathcal{L},\mathcal{G})).$$

Moreover, if $\mathcal{G}$ is injective, then $f^!(\mathcal{L},\mathcal{G})$ is injective.

Proof. By Theorem 6.1 and the remark which follows it, it's enough to show that the functor $f_!(- \otimes_k \mathcal{L})$ is exact, so suppose we have a short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0. \tag{6.1}$$

Since $\mathcal{L}$ is k-flat, the sequence

$$0 \longrightarrow \mathcal{F}' \otimes_k \mathcal{L} \longrightarrow \mathcal{F} \otimes_k \mathcal{L} \longrightarrow \mathcal{F}'' \otimes_k \mathcal{L} \longrightarrow 0$$

remains exact. By Lemma 6.6, $\mathcal{F}' \otimes_k \mathcal{L}$ is soft, and hence $f_!$-acyclic by Theorem 2.9. Therefore, the sequence

$$0 \longrightarrow f_!(\mathcal{F}' \otimes_k \mathcal{L}) \longrightarrow f_!(\mathcal{F} \otimes_k \mathcal{L}) \longrightarrow f_!(\mathcal{F}'' \otimes_k \mathcal{L}) \longrightarrow 0 \tag{6.2}$$

is exact, as required.

To prove the last statement, suppose $\mathcal{G}$ is injective. Then the sequence obtained by applying $\mathrm{Hom}(-,\mathcal{G})$ to (6.2) is exact. But this sequence is isomorphic to the sequence obtained by applying $\mathrm{Hom}(-, f^!(\mathcal{L},\mathcal{G}))$ to (6.1), so $f^!(\mathcal{L},\mathcal{G})$ is injective as well. □

Let $\mathbf{Sh}_{\mathrm{sf}}(X,k)$, respectively $\mathbf{Inj}(X,k)$, denote the full k-linear subcategories of $\mathbf{Sh}(X,k)$ consisting of soft and k-flat, respectively injective, sheaves. The upshot of Proposition 6.7 is that we have a bifunctor

$$f^!(-,-) : \mathbf{Sh}_{\mathrm{sf}}(X,k)^{\mathrm{op}} \times \mathbf{Inj}(Y,k) \longrightarrow \mathbf{Inj}(X,k).$$

Using Construction A.4 (see also Remark A.5), $f^!(-,-)$ induces a bifunctor

$$\mathrm{K}^+(\mathbf{Sh}_{\mathrm{sf}}(X,k))^{\mathrm{op}} \times \mathrm{K}^+(\mathbf{Inj}(Y,k)) \longrightarrow \mathrm{K}^+(\mathbf{Inj}(X,k)). \tag{6.3}$$

If we fix a resolution $\mathcal{K}^\bullet$ of the constant sheaf k in $\mathbf{Sh}(Y,k)$ as in Proposition 6.4, and substitute $\mathcal{K}^\bullet$ into the first slot in (6.3), we get a k-linear functor between the homotopy categories

$$f^!_{\mathcal{K}} : \mathrm{K}^+(\mathbf{Inj}(Y,k)) \longrightarrow \mathrm{K}^+(\mathbf{Inj}(X,k)).$$

Explicitly, if $\mathcal{I}^\bullet \in \mathrm{K}^+(\mathbf{Inj}(Y,k))$ is a complex consisting of injective objects, then $f^!_{\mathcal{K}}(\mathcal{I}^\bullet)$ is the totalisation of the bicomplex

$$(f^!(\mathcal{K}^{-q},\mathcal{I}^p))_{p,q\in\mathbb{Z}}.$$

Now recall that the canonical functors

$$\iota_X : \mathrm{K}^+(\mathbf{Inj}(X,k)) \to \mathrm{D}^+(X,k), \qquad \iota_Y : \mathrm{K}^+(\mathbf{Inj}(Y,k)) \to \mathrm{D}^+(Y,k), \tag{6.4}$$

are equivalences of categories.

Definition 6.8. Suppose X and Y are locally compact spaces of finite cohomological dimension, and $f : X \to Y$ is a continuous map. The *inverse image with compact support*, or the *exceptional inverse image*, of f is the functor

$$f^! := \iota_X \circ f^!_{\mathcal{K}} \circ \iota_Y^{-1} : \mathrm{D}^+(Y,k) \longrightarrow \mathrm{D}^+(X,k),$$

where ι_Y^{-1} is a choice of quasi-inverse for ι_Y.

To state the Verdier duality theorem, we need to recall the Hom complex for sheaves. Using Construction A.4 with the Hom bifunctor

$$\mathrm{Hom}_{\mathbf{Sh}(X,k)}(-,-) : \mathbf{Sh}(X,k)^{\mathrm{op}} \times \mathbf{Sh}(X,k) \longrightarrow k\text{--}\mathbf{Mod}$$

produces a bifunctor

$$\mathrm{Hom}^\bullet(-,-) : \mathrm{K}^+(\mathbf{Sh}(X,k))^{\mathrm{op}} \times \mathrm{K}^+(\mathbf{Sh}(X,k)) \longrightarrow \mathrm{K}(k\text{--}\mathbf{Mod}).$$

If $\mathcal{F}^\bullet, \mathcal{G}^\bullet \in \mathrm{K}^+(\mathbf{Sh}(X,k))$ are complexes of sheaves, then the nth term of the complex $\mathrm{Hom}^\bullet(\mathcal{F}^\bullet,\mathcal{G}^\bullet)$ is

$$\begin{aligned}\mathrm{Hom}^n(\mathcal{F}^\bullet,\mathcal{G}^\bullet) &= \mathrm{Tot}^n\left(\mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^{-q},\mathcal{G}^p)\right)_{p,q\in\mathbb{Z}} \\ &= \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^i,\mathcal{G}^{i+n});\end{aligned}$$

Suppose $\alpha = (\alpha_i) \in \mathrm{Hom}^n(\mathcal{F}^\bullet,\mathcal{G}^\bullet)$. Then the ith component of $d^n\alpha$ is

$$(d^n\alpha)_i = d^{i+n}_{\mathcal{G}} \circ \alpha_i + (-1)^{n+1}\alpha_{i+1} \circ d^i_{\mathcal{F}}.$$

Identifying the complexes $\mathcal{F}^\bullet$ and $\mathcal{G}^\bullet$ with their corresponding graded sheaves, we have

$$\mathrm{Hom}^n(\mathcal{F}^\bullet,\mathcal{G}^\bullet) = \{\text{homogeneous } k\text{-linear maps } \oplus_{i\in\mathbb{Z}}\mathcal{F}^i \to \oplus_{i\in\mathbb{Z}}\mathcal{G}^{i+n}\}.$$

The differential $d^n : \mathrm{Hom}^n(\mathcal{F}^\bullet, \mathcal{G}^\bullet) \to \mathrm{Hom}^{n+1}(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$ can then be described as follows: given a homogeneous map $\alpha : \oplus_{i\in\mathbb{Z}}\mathcal{F}^i \to \oplus_{i\in\mathbb{Z}}\mathcal{G}^{i+n}$, we have

$$d^n\alpha := d_{\mathcal{G}} \circ \alpha - (-1)^n \alpha \circ d_{\mathcal{F}}.$$

Note that this implies

$$\mathrm{Ker}(d^n) = \{\text{maps of complexes } \mathcal{F}^\bullet \to \mathcal{G}^\bullet[n]\},$$

and

$$\mathrm{Im}(d^{n-1}) = \{\text{homotopically trivial maps } \mathcal{F}^\bullet \to \mathcal{G}^\bullet[n]\},$$

so the nth cohomology of the complex $\mathrm{Hom}^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$ consists of morphisms in the homotopy category of sheaf complexes:

$$H^n(\mathrm{Hom}^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)) = \mathrm{Hom}_{\mathrm{K}^+(\mathbf{Sh}(X,k))}(\mathcal{F}^\bullet, \mathcal{G}^\bullet[n]). \tag{6.5}$$

Lemma 6.9. *If $\alpha : \mathcal{F}^\bullet \to \mathcal{G}^\bullet$ is a quasi-isomorphism in $\mathrm{K}^+(\mathbf{Sh}(X,k))$, and $\mathcal{I}^\bullet$ is a complex of injective sheaves, then the natural map*

$$(-) \circ \alpha : \mathrm{Hom}^\bullet(\mathcal{G}^\bullet, \mathcal{I}^\bullet) \longrightarrow \mathrm{Hom}^\bullet(\mathcal{F}^\bullet, \mathcal{I}^\bullet)$$

is a quasi-isomorphism in $\mathrm{K}(k{-}\mathbf{Mod})$.

Proof. By (6.5), $H^n((-) \circ \alpha)$ is given by the top horizontal arrow in the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathrm{K}^+(X,k)}(\mathcal{G}^\bullet, \mathcal{I}^\bullet[n]) & \xrightarrow{(-)\circ\alpha} & \mathrm{Hom}_{\mathrm{K}^+(X,k)}(\mathcal{F}^\bullet, \mathcal{I}^\bullet[n]) \\ \downarrow{\scriptstyle\sim} & & \downarrow{\scriptstyle\sim} \\ \mathrm{Hom}_{\mathrm{D}^+(X,k)}(\mathcal{G}^\bullet, \mathcal{I}^\bullet[n]) & \xrightarrow[\sim]{(-)\circ\alpha} & \mathrm{Hom}_{\mathrm{D}^+(X,k)}(\mathcal{F}^\bullet, \mathcal{I}^\bullet[n]). \end{array}$$

The vertical morphisms are isomorphisms since $\mathcal{I}^\bullet[n]$ consists of injective objects, while the bottom horizontal arrow is an isomorphism since α is a quasi-isomorphism. $\square$

Definition 6.10. The *derived Hom* is the bifunctor

$$R\mathrm{Hom}(-,-) : \mathrm{D}^+(X,k)^{\mathrm{op}} \times \mathrm{D}^+(X,k) \longrightarrow \mathrm{D}(k)$$

defined by

$$R\mathrm{Hom}(-,-) := Q \circ \mathrm{Hom}^\bullet(\iota_X^{-1}(-), \iota_X^{-1}(-)),$$

where ι_X^{-1} is a choice of quasi-inverse to the equivalence of categories ι_X in (6.4), and Q is the canonical functor $\mathrm{K}(k{-}\mathbf{Mod}) \to \mathrm{D}(k{-}\mathbf{Mod})$.

Proposition 6.11. *If $\mathcal{F}^\bullet$ is a fixed complex, then $R\mathrm{Hom}(\mathcal{F}^\bullet, -)$ is the right derived functor of $\mathrm{Hom}^\bullet(\mathcal{F}^\bullet, -)$.*

Proof. Apply Lemma 6.9 to the canonical quasi-isomorphism $\mathcal{F}^\bullet \to \iota_X^{-1}(\mathcal{F}^\bullet)$. $\square$

We can now state and prove the Verdier duality theorem.

Theorem 6.12. *If X and Y have finite cohomological dimension, then there is a natural isomorphism of bifunctors $\mathrm{D}^+(X,k)^{\mathrm{op}} \times \mathrm{D}^+(Y,k) \to \mathrm{D}(k)$:*

$$R\mathrm{Hom}(Rf_!(-), -) \cong R\mathrm{Hom}(-, f^!(-)).$$

Proof. Let $\mathcal{F}^\bullet \in \mathrm{D}^+(X,k)$. By Lemma 6.5, the natural map $\mathcal{F}^\bullet \to \mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet$ is an isomorphism in $\mathrm{D}^+(X,k)$, while by Lemma 6.6, $\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet$ consists of soft sheaves, so we have an isomorphism in $\mathrm{D}^+(Y,k)$:

$$Rf_!\mathcal{F}^\bullet \cong f_!(\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet).$$

We aim to give an isomorphism in $\mathrm{D}(k)$:

$$R\mathrm{Hom}(f_!(\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet), \mathcal{G}^\bullet) \cong R\mathrm{Hom}(\mathcal{F}^\bullet, f^!\mathcal{G}^\bullet),$$

which is natural in both $\mathcal{F}^\bullet$ and $\mathcal{G}^\bullet \in \mathrm{D}^+(Y,k)$. In fact, it is enough to give a natural isomorphism

$$\mathrm{Hom}^\bullet(f_!(\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet), \mathcal{I}^\bullet) \cong \mathrm{Hom}^\bullet(\mathcal{F}^\bullet, f^!_{\mathcal{K}}\mathcal{I}^\bullet), \tag{6.6}$$

where $\mathcal{I}^\bullet := \iota_Y^{-1}(\mathcal{G})$ is the given injective resolution of $\mathcal{G}^\bullet$. This is justified by Proposition 6.11 and the fact that $f^!\mathcal{G}^\bullet = f^!_{\mathcal{K}}\mathcal{I}^\bullet$ consists of injective objects.

The left hand side of (6.6) in degree n is

$$\begin{aligned}
\mathrm{Hom}^n(f_!(\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet), \mathcal{I}^\bullet) &= \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(Y,k)}\left(f_!\left(\oplus_{p+q=i}\mathcal{F}^p \otimes_k \mathcal{K}^q\right), \mathcal{I}^{i+n}\right) \\
&= \prod_{p,q\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(Y,k)}\left(f_!\left(\mathcal{F}^p \otimes_k \mathcal{K}^q\right), \mathcal{I}^{p+q+n}\right),
\end{aligned}$$

while on the right hand side, we have

$$\begin{aligned}
\mathrm{Hom}^n(\mathcal{F}^\bullet, f^!_{\mathcal{K}}\mathcal{I}^\bullet) &= \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(X,k)}\big(\mathcal{F}^i, \mathrm{Tot}^{i+n}(f^!(\mathcal{K}^{-q}, \mathcal{I}^p))_{p,q\in\mathbb{Z}}\big) \\
&= \prod_{i\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(X,k)}\big(\mathcal{F}^i, \oplus_{p+q=i+n} f^!(\mathcal{K}^{-q}, \mathcal{I}^p)\big) \\
&= \prod_{i,j\in\mathbb{Z}} \mathrm{Hom}_{\mathbf{Sh}(X,k)}\big(\mathcal{F}^i, f^!(\mathcal{K}^j, \mathcal{I}^{i+j+n})\big).
\end{aligned}$$

Now for each triple of integers $a, b, n \in \mathbb{Z}$, Proposition 6.7 gives us an isomorphism

$$\mathrm{Hom}_{\mathbf{Sh}(Y,k)}\left(f_!\left(\mathcal{F}^a \otimes_k \mathcal{K}^b\right), \mathcal{I}^{a+b+n}\right) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Sh}(X,k)}\big(\mathcal{F}^a, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n})\big),$$

which we label $\Phi^{a,b,n}$. Therefore, taking products, we get an isomorphism

$$\Phi^n := \textstyle\prod_{a,b\in\mathbb{Z}} \Phi^{a,b,n} : \mathrm{Hom}^n(f_!(\mathcal{F}^\bullet \otimes_k \mathcal{K}^\bullet), \mathcal{I}^\bullet) \cong \mathrm{Hom}^n(\mathcal{F}^\bullet, f^!_{\mathcal{K}}\mathcal{I}^\bullet).$$

It remains to verify that the Φ^n form a morphism of complexes. To do this, observe that the factors $\Phi^{a,b,n}$ are compatible with the differentials on each of the three complexes $\mathcal{F}^\bullet$, $\mathcal{K}^\bullet$ and $\mathcal{I}^\bullet$. More precisely, this means that we have commutative diagrams:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes \mathcal{K}^b), \mathcal{I}^{a+b+n+1}) \arrow[r, "\Phi^{a,b,n+1}"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^a, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n+1})) \\
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^{a+1} \otimes \mathcal{K}^b), \mathcal{I}^{a+b+n+1}) \arrow[r, "\Phi^{a+1,b,n}"] \arrow[u, "(-)\circ f_!(d^a_{\mathcal{F}} \otimes \mathrm{id}_{\mathcal{K}^b})"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^{a+1}, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n+1})), \arrow[u, "(-)\circ d^a_{\mathcal{F}}"']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes \mathcal{K}^b), \mathcal{I}^{a+b+n+1}) \arrow[r, "\Phi^{a,b,n+1}"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^a, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n+1})) \\
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes \mathcal{K}^{b+1}), \mathcal{I}^{a+b+n+1}) \arrow[r, "\Phi^{a,b+1,n}"] \arrow[u, "(-)\circ f_!(\mathrm{id}_{\mathcal{F}^a} \otimes d^b_{\mathcal{K}})"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^a, f^!(\mathcal{K}^{b+1}, \mathcal{I}^{a+b+n+1})), \arrow[u, "f^!(d^b_{\mathcal{K}}, \mathrm{id}_{\mathcal{I}^{a+b+n+1}})\circ(-)"']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes \mathcal{K}^b), \mathcal{I}^{a+b+n+1}) \arrow[r, "\Phi^{a,b,n+1}"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^a, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n+1})) \\
\mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes \mathcal{K}^b), \mathcal{I}^{a+b+n}) \arrow[r, "\Phi^{a,b,n}"] \arrow[u, "d^{a+b+n}_{\mathcal{I}}\circ(-)"] & \mathrm{Hom}_{\mathbf{Sh}(X,k)}(\mathcal{F}^a, f^!(\mathcal{K}^b, \mathcal{I}^{a+b+n})). \arrow[u, "f^!(\mathrm{id}_{\mathcal{K}^b}, d^{a+b+n}_{\mathcal{I}})\circ(-)"']
\end{tikzcd}$$

Let $\alpha = (\alpha_{a,b})_{a,b\in\mathbb{Z}} \in \mathrm{Hom}^n(f_!(\mathcal{F}^\cdot \otimes_k \mathcal{K}^\cdot), \mathcal{I}^\cdot)$, where

$$\alpha_{a,b} \in \mathrm{Hom}_{\mathbf{Sh}(Y,k)}(f_!(\mathcal{F}^a \otimes_k \mathcal{K}^b), \mathcal{I}^{a+b+n}),$$

and let $\beta = \Phi^n(\alpha)$. We have to show that $\Phi^{n+1}(d^n(\alpha)) = d^{n+1}(\beta)$, where d denotes the Hom complex differentials, and it's enough to check that $\Phi^{a,b,n+1}((d^n\alpha)_{a,b}) = (d^n\beta)_{a,b}$. We have

$$\begin{aligned}(d^n\alpha)_{a,b} = d^{a+b+n}_{\mathcal{I}} \circ \alpha_{a,b} + (-1)^{n+1}\alpha_{a+1,b} \circ f_!(d^a_{\mathcal{F}} \otimes \mathrm{id}_{\mathcal{K}_b}) \\ + (-1)^{a+n+1}\alpha_{a,b+1} \circ f_!(\mathrm{id}_{\mathcal{F}_a} \otimes_k d^b_{\mathcal{K}})\end{aligned}$$

and

$$\begin{aligned}(d^n\beta)_{a,b} = f^!(\mathrm{id}_{\mathcal{K}^b}, d^{a+b+n}_{\mathcal{I}}) \circ \beta_{a,b} + (-1)^{a+n+1} f^!(d^b_{\mathcal{K}}, \mathrm{id}_{\mathcal{I}^{a+b+n+1}}) \circ \beta_{a,b+1} \\ + (-1)^{n+1}\beta_{a+1,b} \circ d^a_{\mathcal{F}}.\end{aligned}$$

Now use the commutativity of the diagrams above to complete the proof. □

Corollary 6.13. *The functor $f^!$ is right adjoint to $Rf_!$: for all $\mathcal{F}^\cdot \in \mathrm{D}^+(X,k)$ and $\mathcal{G}^\cdot \in \mathrm{D}^+(Y,k)$, we have a natural isomorphism*

$$\mathrm{Hom}_{\mathrm{D}^+(Y,k)}(Rf_!\mathcal{F}^\cdot, \mathcal{G}^\cdot) \cong \mathrm{Hom}_{\mathrm{D}^+(X,k)}(\mathcal{F}^\cdot, f^!\mathcal{G}^\cdot)$$

Proof. Take $\mathcal{G}^\cdot$ to be a complex of injectives, and using (6.5), apply H^0 to the isomorphism of Theorem 6.12. □

Remark 6.14. Since $f^!$ is the right adjoint of a triangulated functor, it follows that $f^!$ is itself triangulated. See [Stacks, Tag0A8D].

Proposition 6.15. *Suppose X, Y and Z are locally compact spaces of finite cohomological dimension, and $f : X \to Y$ and $g : Y \to Z$ are continuous maps. We have a canonical natural isomorphism $(g \circ f)^! \cong f^! \circ g^!$.*

Proof. The identity follows formally from adjunction and Corollary 3.8. □

Definition 6.16. Let $k \in \mathbf{Sh}(Y, k)$ denote the constant sheaf on Y. The *(relative) dualising complex of f* is defined

$$\mathbb{D}_{X/Y} := f^!(k) \in \mathrm{D}^+(X, k).$$

If Y is a point, then we write $\mathbb{D}_X$ for $\mathbb{D}_{X/Y}$, and we call it the *dualising complex of X*.

In this notation, Theorem 6.12 and Corollary 6.13 say that we have isomorphisms:

$$R\mathrm{Hom}(Rf_!(\mathcal{F}^\bullet), k) \cong R\mathrm{Hom}(\mathcal{F}^\bullet, \mathbb{D}_{X/Y})$$

and

$$\mathrm{Hom}_{\mathrm{D}^+(Y,k)}(Rf_!\mathcal{F}^\bullet, k) \cong \mathrm{Hom}_{\mathrm{D}^+(X,k)}(\mathcal{F}^\bullet, \mathbb{D}_{X/Y}).$$

In the special case where Y is a point, these read

$$R\mathrm{Hom}(R\Gamma_c(X, \mathcal{F}^\bullet), k) \cong R\mathrm{Hom}(\mathcal{F}^\bullet, \mathbb{D}_X)$$

and

$$\mathrm{Hom}_{\mathrm{D}^+(k)}(R\Gamma_c(X, \mathcal{F}^\bullet), k) \cong \mathrm{Hom}_{\mathrm{D}^+(X,k)}(\mathcal{F}^\bullet, \mathbb{D}_X).$$

Finally, we return to the special case where $i : Z \hookrightarrow X$ is the inclusion of a closed subset. In Proposition 6.3, we saw that the pushfoward $i_* = i_!$ has a right adjoint $i^!$. Since $i^!$ is a right adjoint functor, it is left exact, and its right derived functor $Ri^! : \mathrm{D}^+(X, k) \to \mathrm{D}^+(Z, k)$ exists. Also note that $i^!$ preserves injectives, since its left adjoint $i_!$ is exact.

Lemma 6.17. *The right derived functor $Ri^!$ is the right adjoint of $Ri_!$.*

Proof. Let $\mathcal{F}^\bullet \in \mathrm{D}^+(Z, k)$ and $\mathcal{G}^\bullet \in \mathrm{D}^+(X, k)$. We may assume that $\mathcal{F}^\bullet$ and $\mathcal{G}^\bullet$ consist of injective objects, so that $Ri_!\mathcal{F}^\bullet = i_!\mathcal{F}^\bullet$ and $Ri^!\mathcal{G}^\bullet = i^!\mathcal{G}^\bullet$. We then have

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{D}^+(X,k)}(Ri_!\mathcal{F}^\bullet, \mathcal{G}^\bullet) &\cong \mathrm{Hom}_{\mathrm{K}^+(X,k)}(i_!\mathcal{F}^\bullet, \mathcal{G}^\bullet) \\
&\cong \mathrm{Hom}_{\mathrm{K}^+(Z,k)}(\mathcal{F}^\bullet, i^!\mathcal{G}^\bullet) \\
&\cong \mathrm{Hom}_{\mathrm{D}^+(Z,k)}(\mathcal{F}^\bullet, Ri^!\mathcal{G}^\bullet),
\end{aligned}$$

where the last isomorphism follows from the fact that $i^!$ preserves injectives. □

Hence, $Ri^!$ is canonically identified with the functor $i^! : \mathrm{D}^+(X, k) \to \mathrm{D}^+(Z, k)$ of Definition 6.8. This implies a formula for the dualising sheaf $\mathbb{D}_Z$:

Proposition 6.18. *We have $\mathbb{D}_Z \cong Ri^!(\mathbb{D}_X)$.*

Proof. Use Proposition 6.15, taking Z to be a point. □

7. Cohomology and homology with constant coefficients

We begin by reviewing the cup product structure on cohomology. For the moment, X is just a topological space. If we identify the global sections functor $\Gamma(X,-)$ with $\mathrm{Hom}_{\mathbf{Sh}(X,k)}(k,-)$, and then evaluate the corresponding higher derived functors at the constant sheaf k, we get an isomorphism of graded k-modules

$$H^\cdot(X,k) := \oplus_{i\geq 0} H^i(X,k) \cong \oplus_{i\geq 0}\mathrm{Ext}^i_k(k,k).$$

Using the isomorphisms $\mathrm{Ext}^i_k(k,k)\cong \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[i])$, composition in the derived category endows $H^\cdot(X,k)$ with the structure of an associative graded k-algebra:

Definition 7.1. The *cup product* is the graded multiplication

$$(-\cup -) : H^\cdot(X,k)\times H^\cdot(X,k) \longrightarrow H^\cdot(X,k)$$

which is defined as follows. Given cohomology classes $\alpha\in H^i(X,k)$ and $\beta\in H^j(X,k)$, let $\tilde\alpha\in \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[i])$ and $\tilde\beta\in\mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[j])$ be the corresponding morphisms in $\mathrm{D}^+(X,k)$. Then $\alpha\cup\beta\in H^{i+j}(X,k)$ is defined to be the class corresponding to the composition

$$(\tilde\beta[i])\circ\tilde\alpha \in \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[i+j]).$$

The cup product structure on $H^\cdot(X,k)$ is graded commutative. To prove this, we will need to make use of the derived tensor product

$$(-\otimes^L_k -) : \mathrm{D}^-(X,k)\times\mathrm{D}^-(X,k)\longrightarrow \mathrm{D}^-(X,k).$$

This functor exists because k-flat resolutions exist in $\mathrm{D}^-(X,k)$ (see the proof of Lemma 6.6), and if $\mathcal{F}^\cdot,\mathcal{G}^\cdot\in \mathrm{K}^-(X,k)$ are complexes such that $\mathcal{F}^\cdot$ is exact and $\mathcal{G}^\cdot$ consists of k-flats, then $\mathcal{F}^\cdot\otimes_k\mathcal{G}^\cdot$ is exact, by a spectral sequence argument.

Proposition 7.2. *The cup product makes $H^\cdot(X,k)$ into graded commutative algebra: we have*

$$\alpha\cup\beta = (-1)^{ij}\beta\cup\alpha$$

for all $\alpha\in H^i(X,k)$ and $\beta\in H^j(X,k)$.

Proof. Let $\varphi\in\mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[i])$ and $\psi\in\mathrm{Hom}_{\mathrm{D}^+(X,k)}(k,k[j])$ be the morphisms in $\mathrm{D}^+(X,k)$ corresponding to α and β respectively. To compute the compositions $\psi[i]\circ\varphi$ and $\varphi[j]\circ\psi$, we identify $\mathrm{D}^-(X,k)$ with the homotopy category of bounded above k-flat complexes, localised at the class of quasi-isomorphisms. Then φ and ψ are represented by roofs

$$\begin{tikzcd} & \mathcal{F}^\cdot \arrow[dl, "s"'] \arrow[dr, "f"] & \\ k & & k[i] \end{tikzcd} \qquad \begin{tikzcd} & \mathcal{G}^\cdot \arrow[dl, "t"'] \arrow[dr, "g"] & \\ k & & k[j], \end{tikzcd}$$

where $\mathcal{F}^\cdot$ and $\mathcal{G}^\cdot$ are complexes of k-flat sheaves, and s and t are quasi-isomorphisms. Now using the fact that the shift functor $[1]$ is canonically identified with $k[1]\otimes_k(-)$

(with no intervention of sign!), we see that $\psi[i] \circ \varphi$ is computed via the diagram

$$\begin{tikzcd}
 & & \mathcal{F}^\bullet \otimes_k \mathcal{G}^\bullet \arrow[dl, "\mathrm{id}_{\mathcal{F}} \otimes t"'] \arrow[dr, "f \otimes \mathrm{id}_{\mathcal{G}}"] & & \\
 & \mathcal{F}^\bullet \otimes_k k \arrow[dl, "s \otimes \mathrm{id}_k"'] \arrow[dr, "f \otimes \mathrm{id}_k"] & & k[i] \otimes_k \mathcal{G}^\bullet \arrow[dl, "\mathrm{id}_{k[i]} \otimes t"'] \arrow[dr, "\mathrm{id}_{k[i]} \otimes g"] & \\
k \otimes_k k & & k[i] \otimes_k k & & k[i] \otimes_k k[j];
\end{tikzcd}$$

note that $\mathrm{id}_{\mathcal{F}} \otimes t$ is a quasi-isomorphism since $\mathcal{F}^\bullet$ consists of k-flats. On the other hand, $\varphi[j] \circ \psi$ is computed via the diagram

$$\begin{tikzcd}
 & & \mathcal{G}^\bullet \otimes_k \mathcal{F}^\bullet \arrow[dl, "\mathrm{id}_{\mathcal{G}} \otimes s"'] \arrow[dr, "g \otimes \mathrm{id}_{\mathcal{F}}"] & & \\
 & \mathcal{G}^\bullet \otimes_k k \arrow[dl, "t \otimes \mathrm{id}_k"'] \arrow[dr, "g \otimes \mathrm{id}_k"] & & k[j] \otimes_k \mathcal{F}^\bullet \arrow[dl, "\mathrm{id}_{k[j]} \otimes s"'] \arrow[dr, "\mathrm{id}_{k[j]} \otimes f"] & \\
k \otimes_k k & & k[j] \otimes_k k & & k[j] \otimes_k k[i].
\end{tikzcd}$$

The fact that $\psi[i] \circ \varphi = (-1)^{ij} \varphi[j] \circ \psi$ then follows from the commutativity of the two squares:

$$\begin{tikzcd}
\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet \arrow[r, "\mathrm{can}", "\sim"'] \arrow[d, "s \otimes t"] & \mathcal{G}^\bullet \otimes \mathcal{F}^\bullet \arrow[d, "t \otimes s"] \\
k \otimes_k k \arrow[r, "1"] & k \otimes_k k
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{F}^\bullet \otimes \mathcal{G}^\bullet \arrow[r, "\mathrm{can}", "\sim"'] \arrow[d, "f \otimes g"] & \mathcal{G}^\bullet \otimes \mathcal{F}^\bullet \arrow[d, "g \otimes f"] \\
k[i] \otimes_k k[j] \arrow[r, "(-1)^{ij}"] & k[j] \otimes_k k[i]
\end{tikzcd},$$

where can is the canonical isomorphism of Lemma A.6. $\square$

If $f : X \to Y$ is a continuous map, there is a pullback operation on cohomology:

$$f^* : H^i(Y, k) \longrightarrow H^i(X, k), \tag{7.1}$$

which is defined in the following way. If we apply $\Gamma(Y, -)$ to the unit natural transformation $\mathrm{id} \to f_* f^*$, we get a natural transformation $\Gamma(Y, -) \to \Gamma(X, f^*(-))$, and hence a transformation between the derived functors $R\Gamma(Y, -) \to R\Gamma(X, f^*(-))$. Evaluating at the constant sheaf k_Y and taking cohomology then gives us (7.1). Alternatively, and perhaps more elegantly, since $Rf^* = f^*$ by exactness, we have $Rf^* k_Y = f^* k_Y = k_X$ in $\mathrm{D}^+(X, k)$, and (7.1) can be described as

$$f^* : \mathrm{Hom}_{\mathrm{D}^+(Y,k)}(k_Y, k_Y[i]) \longrightarrow \mathrm{Hom}_{\mathrm{D}^+(X,k)}(f^* k_Y, f^* k_Y[i]).$$

This alternative description makes the compatibility of f^* with the cup product obvious:

Proposition 7.3. *If $f : X \to Y$ is a continuous map, then the pullback*

$$f^* : H^\bullet(Y, k) \longrightarrow H^\bullet(X, k)$$

is a morphism of graded k-algebras: given $\alpha \in H^i(Y, k)$ and $\beta \in H^j(Y, k)$, we have

$$f^*(\alpha \cup \beta) = f^* \alpha \cup f^* \beta$$

in $H^{i+j}(X, k)$.

Proof. Immediate from the functoriality of $f^* : \mathrm{D}^+(Y,k) \to \mathrm{D}^+(X,k)$. $\square$

In this way, cohomology is a contravariant functor from the category of topological spaces to the category of graded k-algebras.

We now return to the locally compact setting, so that we can introduce Borel-Moore homology.

Definition 7.4. Suppose X is a locally compact space, and $i \in \mathbb{Z}$. We define the ith *Borel-Moore homology group* with coefficients in k to be the k-module

$$H_i^{BM}(X,k) := H^{-i}\big(R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k)\big).$$

Example 7.5. Suppose k is a field. Then $R\Gamma_{\mathrm{c}}(X,k) \cong \oplus_{i\in\mathbb{Z}} H_{\mathrm{c}}^i(X,k)[-i]$, since every complex of k-vector spaces is quasi-isomorphic to its cohomology. Moreover, k is injective, so we have

$$\begin{aligned} R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k) &\cong \mathrm{Hom}^{\bullet}\left(\oplus_{i\in\mathbb{Z}} H_{\mathrm{c}}^i(X,k)[-i],k\right) \\ &= \oplus_{i\in\mathbb{Z}} \mathrm{Hom}_k(H_{\mathrm{c}}^{-i}(X,k),k)[-i]. \end{aligned}$$

Therefore, $H_i^{BM}(X,k)$ is just the k-linear dual of the vector space $H_{\mathrm{c}}^i(X,k)$.

Remark 7.6. By Verdier duality, we have

$$\begin{aligned} R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k) &\cong R\mathrm{Hom}(k_X,\mathbb{D}_X) \\ &= \mathrm{Hom}^{\bullet}(k_X,\mathbb{D}_X) \\ &= \Gamma(X,\mathbb{D}_X), \end{aligned}$$

so $H_i^{BM}(X,k)$ could alternatively be defined as the $-i$th hypercohomology of the dualising complex $\mathbb{D}_X$.

Suppose $f : X \to Y$ is a proper map of locally compact spaces. Since f is proper, we get an induced map $R\Gamma_{\mathrm{c}}(Y,k) \to R\Gamma_{\mathrm{c}}(X,k)$. Applying $R\mathrm{Hom}(-,k)$ and taking cohomology then gives us a pushforward map

$$f_* : H_i^{BM}(X,k) \longrightarrow H_i^{BM}(Y,k).$$

If $g : Y \to Z$ is a second proper map of locally compact spaces, then it is clear that $(g\circ f)_* = g_* \circ f_*$. In this way, Borel-Moore homology is a covariant functor from the category of locally compact spaces with proper maps to the category of k-modules.

Lemma 7.7. *Suppose X is a finite dimensional locally compact space. Then we have $H_i^{BM}(X,k) = 0$ for all $i > \dim_{\mathrm{c}}(X)$.*

Proof. Choose an isomorphism $R\Gamma_{\mathrm{c}}(X,k) \cong P^{\bullet}$ in $\mathrm{D}^+(k)$, where $P^{\bullet}$ is a complex of projective k-modules with $P^i = 0$ for $i > \dim_{\mathrm{c}}(X)$. By an analogue of Proposition 6.11, we have

$$R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k) \cong R\mathrm{Hom}(P^{\bullet},k) \cong \mathrm{Hom}^{\bullet}(P^{\bullet},k),$$

which implies the result, since $\mathrm{Hom}^{-i}(P^{\bullet},k) = \mathrm{Hom}_k(P^i,k) = 0$ for $i > \dim_{\mathrm{c}}(X)$. $\square$

The Borel-Moore homology groups are acted on by cohomology classes:

Definition 7.8. The *cap product* is the k-bilinear operation

$$(-\cap -) : H_i^{BM}(X,k) \times H^j(X,k) \longrightarrow H_{i-j}^{BM}(X,k)$$

defined as follows. Let $\alpha \in H^j(X,k)$ be a cohomology class, thought of as a map $k_X[-j] \to k_X$ in $\mathrm{D}(X,k)$. Suppose we use Verdier duality to identify $H_i^{BM}(X,k)$ with

$$H^{-i}(R\mathrm{Hom}(k_X, \mathbb{D}_X)) = \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k_X, \mathbb{D}_X[-i]),$$

as in Remark 7.6. If $\tilde{\eta} \in \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k_X, \mathbb{D}_X[-i])$ is the map corresponding to a Borel-Moore homology class $\eta \in H_i^{BM}(X,k)$, then $\eta \cap \alpha$ is defined to be the class in $H_{i-j}^{BM}(X,k)$ corresponding to $\tilde{\eta} \circ \alpha$.

Remark 7.9. The cap product can be also be described without using Verdier duality. Let $\alpha \in \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k_X[-j], k_X)$ be a cohomology class. If we apply the functor $R\Gamma_{\mathrm{c}}(X,-)$, we obtain a map

$$R\Gamma_{\mathrm{c}}(X,k)[-j] = R\Gamma_{\mathrm{c}}(X,k[-j]) \longrightarrow R\Gamma_{\mathrm{c}}(X,k).$$

If we then apply $R\mathrm{Hom}(-,k)$, we get a map

$$R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k) \longrightarrow R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k)[-j],k) \cong R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(X,k),k)[j].$$

The operator $(-) \cap \alpha$ on $H_i^{BM}(X,k)$ is then equal to the $-i$th cohomology of this map.

Proposition 7.10. *The cap product makes the graded k-module $H_\bullet^{BM}(X,k)$ into a right graded $H^\bullet(X,k)$-module.*

Proof. We have to check that $\eta \cap (\alpha \cup \beta) = (\eta \cap \alpha) \cap \beta$, but this is immediate from the definition. □

Now suppose $f : X \to Y$ is a proper map of locally compact spaces. We then have the following 'projection formula':

Proposition 7.11. *If $\eta \in H_i^{BM}(X,k)$ and $\alpha \in H^j(Y,k)$, then we have*

$$f_*(\eta \cap f^*\alpha) = f_*\eta \cap \alpha$$

in $H_{i-j}^{BM}(Y,k)$.

Proof. We use Remark 7.9. Consider the natural transformation of derived functors

$$R\Gamma_c(Y,-) \longrightarrow R\Gamma_c(X, f^*(-)).$$

If we apply this transformation to a morphism $\alpha : k[-j] \to k$ in $\mathrm{D}(Y,k)$, we get a commutative square

$$\begin{tikzcd} R\Gamma_{\mathrm{c}}(Y,k)[-j] \arrow[r, "R\Gamma_{\mathrm{c}}(Y,\alpha)"] \arrow[d] & R\Gamma_{\mathrm{c}}(Y,k) \arrow[d] \\ R\Gamma_{\mathrm{c}}(X,k)[-j] \arrow[r, "R\Gamma_{\mathrm{c}}(X,f^*\alpha)"] & R\Gamma_{\mathrm{c}}(X,k). \end{tikzcd}$$

By unravelling the definitions, one checks that the operator $f_*(-) \cap \alpha$ is obtained by applying $H^{-i}(R\mathrm{Hom}(-,k))$ to the composition

$$R\Gamma_{\mathrm{c}}(Y,k)[-j] \xrightarrow{R\Gamma_{\mathrm{c}}(Y,\alpha)} R\Gamma_{\mathrm{c}}(Y,k) \longrightarrow R\Gamma_{\mathrm{c}}(X,k),$$

while $f_*((-) \cap f^*\alpha)$ is obtained by applying the same functor to

$$R\Gamma_{\mathrm{c}}(Y,k)[-j] \longrightarrow R\Gamma_{\mathrm{c}}(X,k)[-j] \xrightarrow{R\Gamma_{\mathrm{c}}(X,f^*\alpha)} R\Gamma_{\mathrm{c}}(X,k).$$

□

8. Orientation and Poincaré duality

We discuss orientations of manifolds, and deduce Poincaré duality from Verdier duality.

Lemma 8.1. *Suppose X is a locally compact space of dimension n. Then then there is a sheaf $\mathscr{or}_{X,k} \in \mathbf{Sh}(X,k)$, with sections*

$$\Gamma(U, \mathscr{or}_{X,k}) = \mathrm{Hom}_k(H^n_{\mathrm{c}}(U,k), k)$$

for any open subset $U \subset X$.

Proof. We show the contravariant functor $\mathcal{G} \mapsto \mathrm{Hom}_k(H^n_{\mathrm{c}}(X,\mathcal{G}), k)$ is representable using Theorem 6.1. The result follows, since the sections of the representing sheaf over an open subset $U \subset X$ are given by

$$\mathrm{Hom}_k(H^n_{\mathrm{c}}(X,k_U), k) = \mathrm{Hom}_k(H^n_{\mathrm{c}}(U,k), k).$$

Recall that colimits in the abelian category $\mathbf{Sh}(X,k)$ are build out of coproducts and cokernels. The functor $\mathcal{G} \mapsto \mathrm{Hom}_k(H^n_{\mathrm{c}}(X,\mathcal{G}), k)$ clearly sends coproducts to products, so we need to show it sends cokernels to kernels. But $\mathrm{Hom}_k(-,k)$ sends cokernels to kernels, so we just have to check that $H^n_{\mathrm{c}}(X,-)$ preserves cokernels.

Suppose $\mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ in an exact sequence in $\mathbf{Sh}(X,k)$. Let $\mathcal{K}$ be the kernel of the map $\mathcal{G} \to \mathcal{H}$, so that we have a factorisation

$$\begin{tikzcd}
\mathcal{K} \arrow[dr, hook] & & & \\
\mathcal{F} \arrow[u, two heads] \arrow[r] & \mathcal{G} \arrow[r] & \mathcal{H} \arrow[r] & 0.
\end{tikzcd}$$

Now if we take the long exact cohomology sequence associated to the short exact sequence $0 \to \mathcal{K} \to \mathcal{G} \to \mathcal{H} \to 0$, we see that the map $H^n_{\mathrm{c}}(X,\mathcal{G}) \to H^n_{\mathrm{c}}(X,\mathcal{H})$ is surjective (we have $H^{n+1}_{\mathrm{c}}(X,\mathcal{K}) = H^{n+1}_{\mathrm{c}}(X,\mathcal{G}) = 0$ because $\dim_{\mathrm{c}}(X) = n$). By the same reasoning, the map $H^n_{\mathrm{c}}(X,\mathcal{F}) \to H^n_{\mathrm{c}}(X,\mathcal{K})$ is surjective. So we have a factorisation on the level of cohomology:

$$\begin{tikzcd}
H^n_{\mathrm{c}}(X,\mathcal{K}) \arrow[dr] & & & \\
H^n_{\mathrm{c}}(X,\mathcal{F}) \arrow[u, two heads] \arrow[r] & H^n_{\mathrm{c}}(X,\mathcal{G}) \arrow[r] & H^n_{\mathrm{c}}(X,\mathcal{H}) \arrow[r] & 0.
\end{tikzcd}$$

This shows the bottom row is exact, as required. $\square$

Definition 8.2. The sheaf $\mathscr{or}_{X,k}$ of Lemma 8.1 is called the *orientation sheaf* of X relative to k.

Now suppose X is an n-dimensional topological manifold. Then X has a covering by open subspaces homeomorphic to $\mathbb{R}^n$, and Theorem 4.10 implies $\mathscr{or}_{X,k}$ is a locally constant sheaf with stalk k.

Definition 8.3. We say that a topological manifold X is *orientable* relative to k if the locally constant sheaf $\mathscr{or}_{X,k}$ is constant. An isomorphism of k-sheaves $\mathscr{or}_{X,k} \cong k_X$ is called a *k-orientation* of X. If X is orientable relative to k, then we sometimes say that X is *oriented* relative to k, if a choice of k-orientation is understood to have been made, but left implicit.

In particular, if X is orientable relative to the ring $\mathbb{Z}$, then we say simply that X is *orientable*, and we write $\mathscr{or}_X := \mathscr{or}_{X,\mathbb{Z}}$. An *orientation* is an isomorphism of sheaves $\mathscr{or}_X \cong \mathbb{Z}_X$. The following lemma implies that if X is orientable, then X is orientable relative to k for any ring k; moreover, any orientation of X determines a k-orientation by tensoring with k.

Lemma 8.4. *If X is a topological manifold, then we have $\mathscr{or}_{X,k} \cong \mathscr{or}_X \otimes_{\mathbb{Z}} k$.*

Proof. Let a morphism of sheaves $\mathscr{or}_X \otimes_{\mathbb{Z}} k \to \mathscr{or}_{X,k}$ be given over an open subset $U \subset X$ by the natural map

$$\mathrm{Hom}_{\mathbb{Z}}(H^n_{\mathrm{c}}(U,\mathbb{Z}),\mathbb{Z}) \otimes_{\mathbb{Z}} k \longrightarrow \mathrm{Hom}_k(H^n_{\mathrm{c}}(U,k),k). \tag{8.1}$$

(note that $H^n_{\mathrm{c}}(U,\mathbb{Z}) \otimes_{\mathbb{Z}} k \cong H^n_{\mathrm{c}}(U,k)$, by the universal coefficient theorem [Iv, VI.5]). Now if U is homeomorphic to $\mathbb{R}^n$, then (8.1) is an isomorphism by Theorem 4.10, so $\mathscr{or}_X \otimes_{\mathbb{Z}} k \to \mathscr{or}_{X,k}$ is an isomorphism at each of its stalks. □

We are now ready to discuss Poincaré duality for oriented manifolds. The first step is to compute the dualising complex $\mathbb{D}_X$, which turns out, in this very special case, to be nothing more than a shift of the orientation sheaf:

Theorem 8.5. *Suppose X is a topological manifold of dimension n. Then we have an isomorphism in $\mathrm{D}^+(X,k)$:*

$$\mathbb{D}_X \cong \mathscr{or}_{X,k}[n].$$

Proof. We will show that

$$\mathcal{H}^i(\mathbb{D}_X) = \begin{cases} \mathscr{or}_{X,k} & \text{if } i = -n \\ 0 & \text{otherwise,} \end{cases}$$

where $\mathcal{H}^i(\mathbb{D}_X)$ is the ith cohomology sheaf of the complex $\mathbb{D}_X$. The isomorphism $\mathbb{D}_X \cong \mathscr{or}_{X,k}[n]$ is then obtained by using truncation functors.

Suppose $U \subset X$ is an open subset homeomorphic to $\mathbb{R}^n$. By Verdier duality (Theorem 6.12), we have an isomorphism in the derived category of k-modules:

$$R\mathrm{Hom}(k_U,\mathbb{D}_X) \cong R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(U,k),k).$$

Since $\mathbb{D}_X$ consists of injective sheaves (by definition), Proposition 6.11 implies

$$R\mathrm{Hom}(k_U,\mathbb{D}_X) = \mathrm{Hom}^\bullet(k_U,\mathbb{D}_X).$$

On the other hand, by Theorem 4.10, we have an isomorphism in $\mathrm{D}(k)$:

$$R\Gamma_{\mathrm{c}}(U,k) \cong H^n_{\mathrm{c}}(U,k)[-n],$$

and since $H^n_{\mathrm{c}}(U,k)[-n] = k[-n]$ is a complex consists of *projective* k-modules, an analogue of Proposition 6.11 implies

$$R\mathrm{Hom}(R\Gamma_{\mathrm{c}}(U,k),k) \cong \mathrm{Hom}^\bullet(H^n_{\mathrm{c}}(U,k)[-n],k).$$

So altogether, we have an isomorphism in the homotopy category of k-modules:

$$\mathrm{Hom}^\bullet(k_U,\mathbb{D}_X) \cong \mathrm{Hom}^\bullet(H^n_{\mathrm{c}}(U,k)[-n],k).$$

If we compute the Hom complexes, we find that this simplifies to

$$\Gamma(U,\mathbb{D}_X) \cong \mathrm{Hom}_k(H^n_{\mathrm{c}}(U,k),k)[n]$$

But $\mathcal{H}^i(\mathbb{D}_X)$ is the sheaf associated to the presheaf $U \mapsto H^i(\Gamma(U,\mathbb{D}_X))$. Since X is covered by open sets homeomorphic to $\mathbb{R}^n$, this completes the proof. □

Recall that by Verdier duality (Remark 7.6), $H_i^{BM}(X,k)$ can be computed as the $-i$th hypercohomology of the dualising complex. In particular, for $i=n$, we have

$$H_n^{BM}(X,k) \cong H^{-n}(X,\mathbb{D}_X) \cong \mathrm{Hom}_k(k_X, \mathscr{or}_{X,k}). \tag{8.2}$$

Definition 8.6. Suppose X is a topological n-manifold, orientable relative to k, and that $\mu : k_X \cong \mathscr{or}_{X,k}$ is a fixed k-orientation of X. The *fundamental class* of X is the Borel-Moore homology class $[X] \in H_n^{BM}(X,k)$ corresponding to μ under the isomorphism (8.2).

Now that we have defined fundamental classes, we can finally state the Poincaré duality theorem, in something resembling its classical form.

Theorem 8.7. *Let X be a topological n-manifold, oriented relative to k, and let $[X] \in H_n^{BM}(X,k)$ be the fundamental class of X. Then capping with $[X]$ gives an isomorphism*

$$[X] \cap (-) : H^i(X,k) \xrightarrow{\sim} H_{n-i}^{BM}(X,k).$$

Proof. Let $\mu : k_X \cong \mathscr{or}_{X,k}$ be the k-orientation of X corresponding to the choice of fundamental class. By Verdier duality (Theorem 6.12) and Theorem 8.5, we have an isomorphism

$$R\mathrm{Hom}(k_X, \mathscr{or}_{X,k}) \cong R\mathrm{Hom}(\Gamma_c(X,k),k)[-n].$$

Using μ to identify the left hand side of this equation with $R\mathrm{Hom}(k_X,k_X)$ and taking cohomology in degree i, we get an isomorphism

$$\mathrm{Hom}_{\mathrm{D}^+(X,k)}(k_X, k_X[i]) \cong \mathrm{Hom}_{\mathrm{D}^+(X,k)}(k_X, \mathscr{or}_{X,k}[i]) \cong H_{n-i}^{BM}(X,k),$$

which is equal to $[X] \cap (-)$ by definition. □

We end this section with a cohomological criterion for orientability, which makes contact with alternative definitions:

Proposition 8.8. *Suppose X is a connected topological manifold of dimension n. Then the k-orientability of X is decided by its top Borel-Moore homology: we have*

$$H_n^{BM}(X,k) = \begin{cases} k & \text{if } X \text{ is orientable relative to } k \\ 0 & \text{otherwise.} \end{cases}$$

Proof. First suppose X is orientable relative to k. Then by Poincaré duality 8.7, we have

$$H_n^{BM}(X,k) \cong H^0(X,k) = k,$$

since X is connected.

Now suppose $H_n^{BM}(X,k) \neq 0$, so that $\Gamma(X, \mathscr{or}_{X,k}) \neq 0$ by (8.2). Let s be a nonzero global section of $\mathscr{or}_{X,k}$. Since $\mathscr{or}_{X,k}$ is locally constant, $\mathrm{Supp}(s)$ is an open subset of X. On the other hand, $\mathrm{Supp}(s)$ is also closed and nonempty, so we have $\mathrm{Supp}(s) = X$ by the connectedness of X. Therefore, we must have $\mathscr{or}_{X,k} \cong k$, so X is orientable relative to k. □

9. Application to algebraic varieties

Reminder of the definition of algebraic variety: an integral, separated scheme of finite type over $\mathbb{C}$. Explain how the Nullstellensatz endows X with the classical topology.

Proposition 9.1. *A complex algebraic variety X with the classical topology is a locally compact space.*

Proof. Separatedness implies the Hausdorff condition. The 'locally compact' condition is a local question, so we can assume X is affine. Since it holds for $\mathbb{C}^n$ it also holds for its closed subspace X. $\square$

- Prove that if X is an algebraic variety, then $\dim_{\mathrm{c}} X = 2 \dim X$.
- If Z is a codimension d oriented submanifold of an oriented manifold X, then using the local duality isomorphism, Z determines a class $[Z]$ in $H^d_Z(X, A)$.
- Generalise this to the case where Z is a codimension d subvariety of an smooth algebraic variety X.
- We can use Poincaré duality to get classes in BM homology.
- Properties: push and pull formulas, projection formula.
- Comparison of Chow groups to BM homology.
- Computing the BM homology if a paving by affine spaces in known. Example: the flag variety.

Appendix A. Bicomplexes

Let $\mathcal{A}$ be a k-linear category. A *bicomplex* $(C^{\bullet,\bullet}, d_\rightarrow, d_\uparrow)$ in $\mathcal{A}$ is a collection of objects $\{C^{p,q}\}^{p,q\in\mathbb{Z}}$ in $\mathcal{A}$, together with morphisms

$$d^{p,q}_\rightarrow : C^{p,q} \longrightarrow C^{p+1,q}, \qquad d^{p,q}_\uparrow : C^{p,q} \longrightarrow C^{p,q+1},$$

satisfying the identities

$$d^{p+1,q}_\rightarrow d^{p,q}_\rightarrow = 0, \qquad d^{p,q+1}_\uparrow d^{p,q}_\uparrow = 0, \qquad d^{p+1,q}_\uparrow d^{p,q}_\rightarrow + d^{p,q+1}_\rightarrow d^{p,q}_\uparrow = 0.$$

A *morphism* of two bicomplexes $C^{\bullet,\bullet}$ and $D^{\bullet,\bullet}$ is a collection of morphisms in $\mathcal{A}$

$$f^{p,q} : C^{p,q} \longrightarrow D^{p,q}$$

which commute with the differentials, in the sense that

$$d^{p,q}_\rightarrow f^{p,q} = f^{p+1,q} d^{p,q}_\rightarrow, \qquad d^{p,q}_\uparrow f^{p,q} = f^{p,q+1} d^{p,q}_\uparrow$$

for all $p, q \in \mathbb{Z}$. In this way, bicomplexes in $\mathcal{A}$ form a category, which we denote $\mathbf{Bicom}(\mathcal{A})$.

We often wish to assume our bicomplexes are bounded in some way. More precisely, we say that a bicomplex $(C^{\bullet,\bullet}, d_\rightarrow, d_\uparrow)$ is *bounded to the left* or *left-bounded* if there exists an integer p_0 such that for all $p < p_0$ and all $q \in \mathbb{Z}$, we have $C^{p,q} = 0$. The definitions of *right-bounded*, *bounded below* and *bounded above* are similar.

Suppose we are given two morphisms of bicomplexes $f, g : C^{\bullet,\bullet} \to D^{\bullet,\bullet}$. We say that f and g are *homotopic* if there exist morphisms

$$h^{p,q}_\leftarrow : C^{p,q} \longrightarrow D^{p-1,q}, \qquad h^{p,q}_\downarrow : C^{p,q} \longrightarrow D^{p,q-1},$$

such that

$$d_\rightarrow h_\downarrow + h_\downarrow d_\rightarrow = 0, \qquad d_\uparrow h_\leftarrow + h_\leftarrow d_\uparrow = 0,$$

and

$$f - g = d_\rightarrow h_\leftarrow + h_\leftarrow d_\rightarrow + d_\uparrow h_\downarrow + h_\downarrow d_\uparrow,$$

and we refer to the collection of morphisms $(h_\leftarrow, h_\downarrow)$ as a *homotopy*.

Lemma A.1. *Homotopy is an equivalence relation of morphisms of bicomplexes* $C^{\cdot,\cdot} \to D^{\cdot,\cdot}$

Proof. The proof works exactly the same way as for ordinary complexes. □

Hence, we can divide out by homotopy equivalence to form a homotopy category of bicomplexes in $\mathcal{A}$, denoted $\mathrm{K}(\mathbf{Bicom}(\mathcal{A}))$.

The *totalisation* of a bicomplex $(C^{\cdot,\cdot}, d_{\rightarrow}, d_{\uparrow}) \in \mathbf{Bicom}(\mathcal{A})$ is the complex $\mathrm{Tot}(C^{\cdot,\cdot})$ in $\mathcal{A}$ with nth term

$$\mathrm{Tot}^n(C^{\cdot,\cdot}) := \prod_{p+q=n} C^{p,q};$$

the differential $d^n : \mathrm{Tot}^n(C^{\cdot,\cdot}) \to \mathrm{Tot}^{n+1}(C^{\cdot,\cdot})$ is defined on the summand $C^{p,q}$ by

$$d^n|_{C^{p,q}} := d_{\rightarrow}^{p,q} + d_{\uparrow}^{p,q}.$$

Note that we have

$$\begin{aligned} d^2 &= (d_{\rightarrow} + d_{\uparrow})(d_{\rightarrow} + d_{\uparrow}) \\ &= d_{\rightarrow}^2 + d_{\rightarrow}d_{\uparrow} + d_{\uparrow}d_{\rightarrow} + d_{\uparrow}^2 \\ &= 0, \end{aligned}$$

by the definition of bicomplex.

Remark A.2. Given that we only assumed $\mathcal{A}$ was additive, the totalisation may not exist if $\mathcal{A}$ does not have arbitrary products. In our applications, $\mathcal{A}$ will either be an abelian category (in which case infinite products do exist), or else we will impose some boundedness condition on our bicomplexes to ensure that the product $\mathrm{Tot}^n(C^{\cdot,\cdot})$ is finite. (In this case, it is better to write $\mathrm{Tot}^n(C^{\cdot,\cdot})$ as a direct sum.)

Suppose $\mathcal{A}$ is such that all totalisations in $\mathbf{Bicom}(\mathcal{A})$ exist. To every morphism of bicomplexes $f : C^{\cdot,\cdot} \to D^{\cdot,\cdot}$, we can, in the obvious way, associate a morphism on the corresponding totalisations:

$$\mathrm{Tot}(f) : \mathrm{Tot}(C^{\cdot,\cdot}) \longrightarrow \mathrm{Tot}(D^{\cdot,\cdot}).$$

So we have a functor

$$\mathrm{Tot} : \mathbf{Bicom}(\mathcal{A}) \longrightarrow \mathbf{Com}(\mathcal{A}).$$

Next, we show that totalisation preserves homotopy equivalence of morphisms:

Lemma A.3. *Suppose* $f, g : C^{\cdot,\cdot} \to D^{\cdot,\cdot}$ *are two morphisms of bicomplexes, where* $C^{\cdot,\cdot}, D^{\cdot,\cdot} \in \mathbf{Bicom}(\mathcal{A})$*. If* f *and* g *are homotopy equivalent, then* $\mathrm{Tot}(f)$ *and* $\mathrm{Tot}(g)$ *are homotopy equivalent in* $\mathbf{Com}(\mathcal{A})$*.*

Proof. Let $(h_{\leftarrow}, h_{\downarrow})$ be a homotopy between f and g. For each $n \in \mathbb{Z}$, define a morphism $h^n : \mathrm{Tot}^n(C^{\cdot,\cdot}) \to \mathrm{Tot}^{n-1}(D^{\cdot,\cdot})$ by setting

$$h^n|_{C^{p,q}} := h_{\leftarrow}^{p,q} + h_{\downarrow}^{p,q}$$

for all $p, q \in \mathbb{Z}$ such that $p + q = n$. We then have to check that

$$\mathrm{Tot}^n(f) - \mathrm{Tot}^n(g) = d^{n-1}h^n + h^{n+1}d^n.$$

This can be done simply by choosing an element $x \in C^{p,q}$ (which is legitimate in the context of our applications), and writing everything out. □

As a consequence of Lemma A.3, totalisation descends to a k-linear functor

$$\mathrm{Tot} : \mathrm{K}(\mathbf{Bicom}(\mathcal{A})) \longrightarrow \mathrm{K}(\mathcal{A}). \tag{A.1}$$

Construction A.4. Let $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ be k-linear categories. Suppose we are given a k-linear bifunctor

$$\beta : \mathcal{A}^{\mathrm{op}} \times \mathcal{B} \longrightarrow \mathcal{C}.$$

Our goal is to construct from β a k-linear bifunctor between the homotopy categories

$$\mathrm{K}(\mathcal{A})^{\mathrm{op}} \times \mathrm{K}(\mathcal{B}) \longrightarrow \mathrm{K}(\mathcal{C}).$$

As a first step, we construct a bifunctor

$$\beta^{\cdot,\cdot} : \mathbf{Com}(\mathcal{A})^{\mathrm{op}} \times \mathbf{Com}(\mathcal{B}) \longrightarrow \mathbf{Bicom}(\mathcal{C}).$$

Given complexes $A^{\cdot} \in \mathbf{Com}(\mathcal{A})$ and $B^{\cdot} \in \mathbf{Com}(\mathcal{B})$, we define a bicomplex $\beta^{\cdot,\cdot}(A^{\cdot}, B^{\cdot})$ by setting

$$\beta^{p,q}(A^{\cdot}, B^{\cdot}) := \beta(A^{-q}, B^{p})$$

and

$$d_{\rightarrow}^{p,q} := \beta(\mathrm{id}_{A^{-q}}, d_B^{p}), \qquad d_{\uparrow}^{p,q} := (-1)^{p+q+1}\beta(d_A^{-q-1}, \mathrm{id}_{B^{p}}).$$

If $(f : A'^{\cdot} \to A^{\cdot}, g : B^{\cdot} \to B'^{\cdot})$ is a morphism in $\mathbf{Com}(\mathcal{A})^{\mathrm{op}} \times \mathbf{Com}(\mathcal{B})$, setting

$$\beta^{p,q}(f, g) := \beta(f^{-q}, g^{p}) : \beta^{p,q}(A^{\cdot}, B^{\cdot}) \longrightarrow \beta^{p,q}(A'^{\cdot}, B'^{\cdot})$$

defines a morphism of bicomplexes $\beta^{\cdot,\cdot}(f, g) : \beta^{\cdot,\cdot}(A^{\cdot}, B^{\cdot}) \to \beta^{\cdot,\cdot}(A'^{\cdot}, B'^{\cdot})$. It is clear that $\beta^{\cdot,\cdot}$ is a k-linear bifunctor, since β is.

Next, we show that $\beta^{\cdot,\cdot}$ preserves homotopy equivalence classes of morphisms, and hence descends to a bifunctor

$$\beta^{\cdot,\cdot} : \mathrm{K}(\mathcal{A})^{\mathrm{op}} \times \mathrm{K}(\mathcal{B}) \longrightarrow \mathrm{K}(\mathbf{Bicom}(\mathcal{C})). \tag{A.2}$$

We can check this separately in each variable. Fix a complex $A^{\cdot} \in \mathbf{Com}(\mathcal{A})$, and suppose $f, g : B^{\cdot} \to B'^{\cdot}$ are homotopic morphisms in $\mathbf{Com}(\mathcal{B})$. Let h is a homotopy between f and g, and define

$$h_{\leftarrow}^{p,q} := \beta(\mathrm{id}_{A^{-q}}, h^{p}), \qquad h_{\downarrow}^{p,q} = 0$$

for all $p, q \in \mathbb{Z}$. We claim that $(h_{\leftarrow}, h_{\downarrow})$ is a homotopy between $\beta^{\cdot,\cdot}(\mathrm{id}_A, f)$ and $\beta^{\cdot,\cdot}(\mathrm{id}_A, g)$. Since $h_{\downarrow} = 0$, it's clear that $d_{\rightarrow}h_{\downarrow} + h_{\downarrow}d_{\rightarrow} = 0$. The other identity $d_{\uparrow}h_{\leftarrow} + h_{\leftarrow}d_{\uparrow} = 0$ follows from the bifunctoriality of β and the sign in the definition of $d_{\uparrow}$. Finally, it is straightforward to check that

$$\beta^{p,q}(\mathrm{id}_A, f) - \beta^{p,q}(\mathrm{id}_A, g) = d_{\rightarrow}^{p-1,q}h_{\leftarrow}^{p,q} + h_{\leftarrow}^{p+1,q}d_{\rightarrow}^{p,q}.$$

Therefore, $\beta^{\cdot,\cdot}(\mathrm{id}_A, f)$ and $\beta^{\cdot,\cdot}(\mathrm{id}_A, g)$ are homotopic morphisms of bicomplexes. If $B^{\cdot} \in \mathbf{Com}(\mathcal{B})$ is fixed, then the verification that $\beta^{\cdot,\cdot}(-, B^{\cdot})$ preserves homotopies is similar.

Finally, use (A.1) and (A.2) to define a composite functor

$$\mathrm{K}(\mathcal{A})^{\mathrm{op}} \times \mathrm{K}(\mathcal{B}) \xrightarrow{\beta^{\cdot,\cdot}} \mathrm{K}(\mathbf{Bicom}(\mathcal{C})) \xrightarrow{\mathrm{Tot}} \mathrm{K}(\mathcal{C}). \tag{A.3}$$

Remark A.5. Denote by $\mathrm{K}^{+,+}(\mathbf{Bicom}(\mathcal{C}))$ the full subcategory of $\mathrm{K}(\mathbf{Bicom}(\mathcal{C}))$ consisting of bicomplexes which are bounded to the left and bounded below. It is easy to see that the functor (A.3) restricts to a bifunctor

$$\mathrm{K}^{-}(\mathcal{A})^{\mathrm{op}} \times \mathrm{K}^{+}(\mathcal{B}) \xrightarrow{\beta^{\cdot,\cdot}} \mathrm{K}^{+,+}(\mathbf{Bicom}(\mathcal{C})) \xrightarrow{\mathrm{Tot}} \mathrm{K}^{+}(\mathcal{C}).$$

Finally, we discuss tensor products of complexes. They can be defined using a method similar to that of Construction A.4, but we will treat them separately, since we use a different sign convention for the differentials.

Let $(\mathcal{A}, \otimes, I)$ be a k-linear symmetric monoidal category. We have in mind some category of k-sheaves on a space X, where $\otimes$ is the usual tensor product for sheaves, and the unit I is the constant sheaf k_X. Given two complexes $A^{\cdot}, B^{\cdot} \in \mathbf{Com}^*(\mathcal{A})$, where $* \in \{+, -, \mathrm{b}\}$, we define their *tensor product* to be the complex $A^{\cdot} \otimes B^{\cdot}$ with nth term

$$(A^{\cdot} \otimes B^{\cdot})^n := \bigoplus_{p+q=n} A^p \otimes B^q,$$

and differential $d^n : (A^{\cdot} \otimes B^{\cdot})^n \to (A^{\cdot} \otimes B^{\cdot})^{n+1}$ given by

$$d^n|_{A^p \otimes B^q} := d_A^p \otimes \mathrm{id}_{B_q} + (-1)^p \,\mathrm{id}_{A^p} \otimes d_B^q.$$

In other words, $A^{\cdot} \otimes B^{\cdot}$ is the totalisation of the bicomplex

$$\begin{tikzcd}
 & \vdots & \vdots & \\
\cdots \arrow[r] & A^p \otimes B^{q+1} \arrow[u] \arrow[r, "d_A^p \otimes \mathrm{id}_{B_{q+1}}"] & A^{p+1} \otimes B^{q+1} \arrow[u] \arrow[r] & \cdots \\
\cdots \arrow[r] & A^p \otimes B^q \arrow[u, "(-1)^p \mathrm{id}_{A^p} \otimes d_B^q"] \arrow[r, "d_A^p \otimes \mathrm{id}_{B_q}"] & A^{p+1} \otimes B^q \arrow[u, "(-1)^{p+1} \mathrm{id}_{A^{p+1}} \otimes d_B^q"'] \arrow[r] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

If $f : A^{\cdot} \to A'^{\cdot}$ and $g : B^{\cdot} \to B'^{\cdot}$ are morphisms of complexes, then we define $f \otimes g$ to be the totalisation of the morphism of bicomplexes $(f, g) : (A^p \otimes B^q) \to (A'^p \otimes B'^q)$ which is given pointwise by $f^p \otimes g^q$. It follows that the tensor product is a bifunctor

$$(- \otimes -) : \mathrm{K}^*(\mathcal{A}) \times \mathrm{K}^*(\mathcal{A}) \longrightarrow \mathrm{K}^*(\mathcal{A}).$$

Lemma A.6. *We have a canonical isomorphism $A^{\cdot} \otimes B^{\cdot} \cong B^{\cdot} \otimes A^{\cdot}$, induced from the morphism of bicomplexes which is given pointwise by*

$$(-1)^{pq} : A^p \otimes B^q \xrightarrow{\sim} B^q \otimes A^p.$$